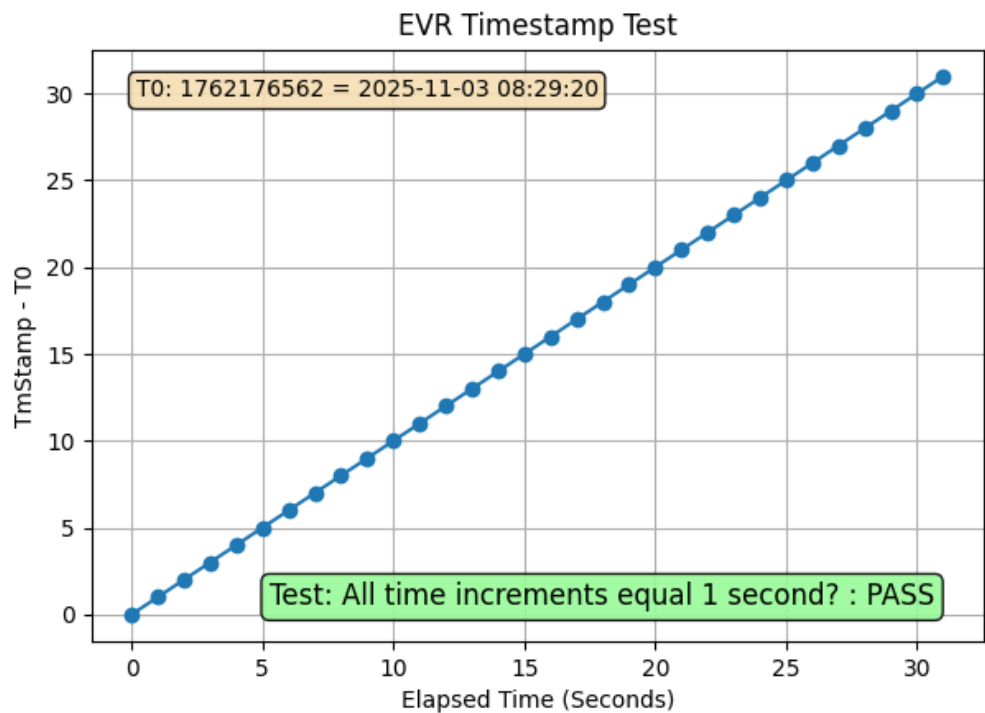


PSC Functional Test Report

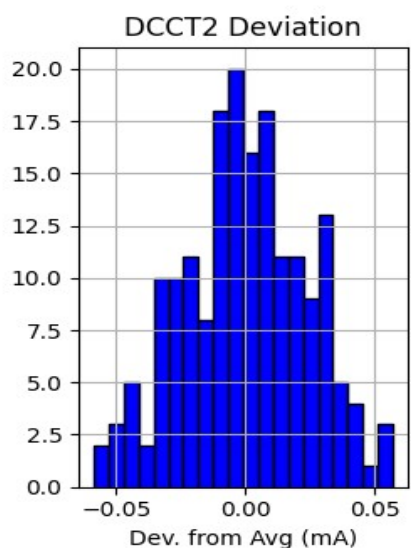
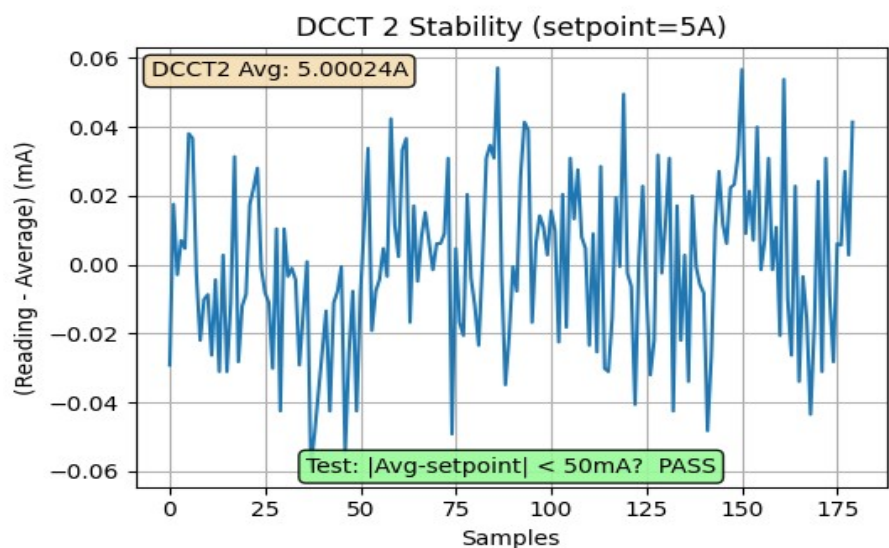
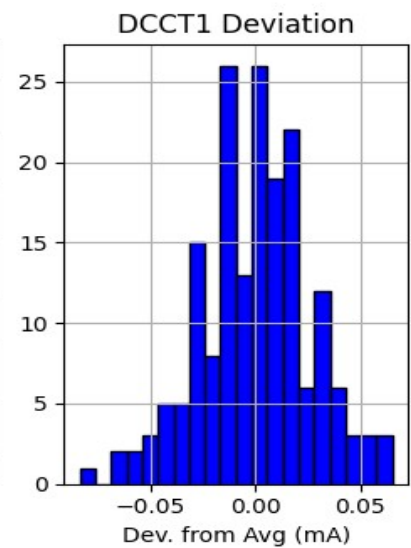
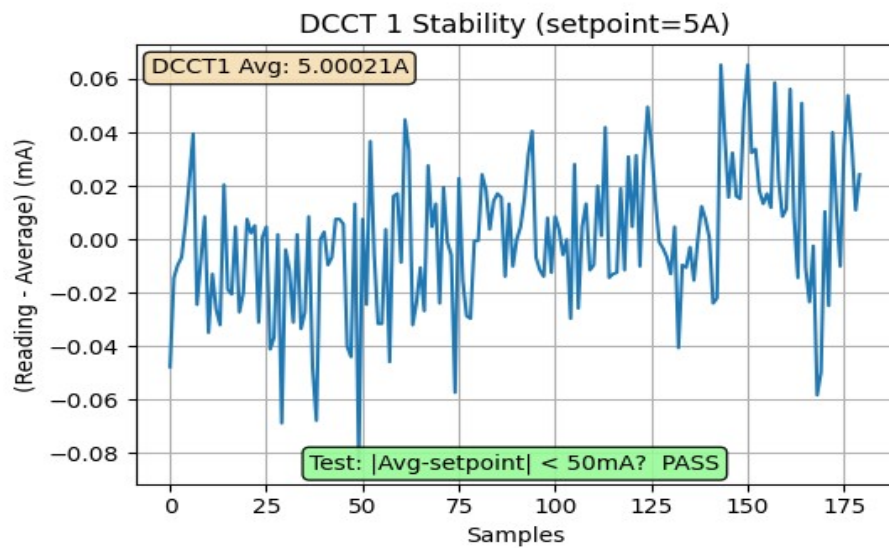
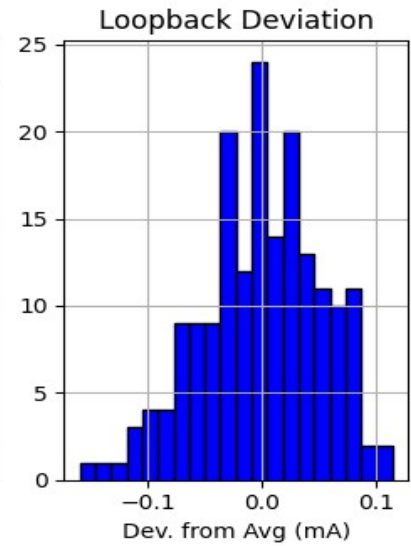
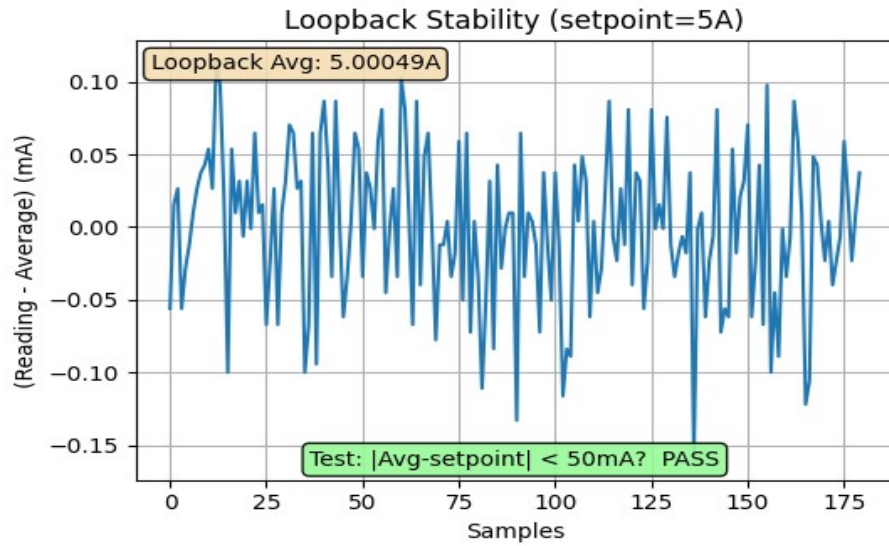
PSC Functional Test Results	
Power Supply Controller Configuration	
Description	PSC-4CH-HSF-EIC
Serial Number	0001
Number of Channels	4
Resolution	HS (20bit)
Bandwidth	F
Channel 1 Polarity	Unipolar
Channel 2 Polarity	Unipolar
Channel 3 Polarity	Unipolar
Channel 4 Polarity	Unipolar



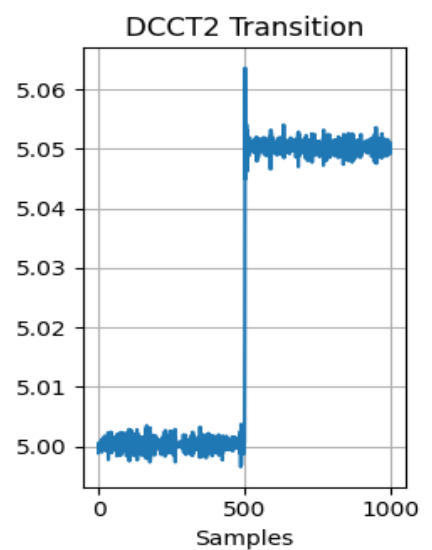
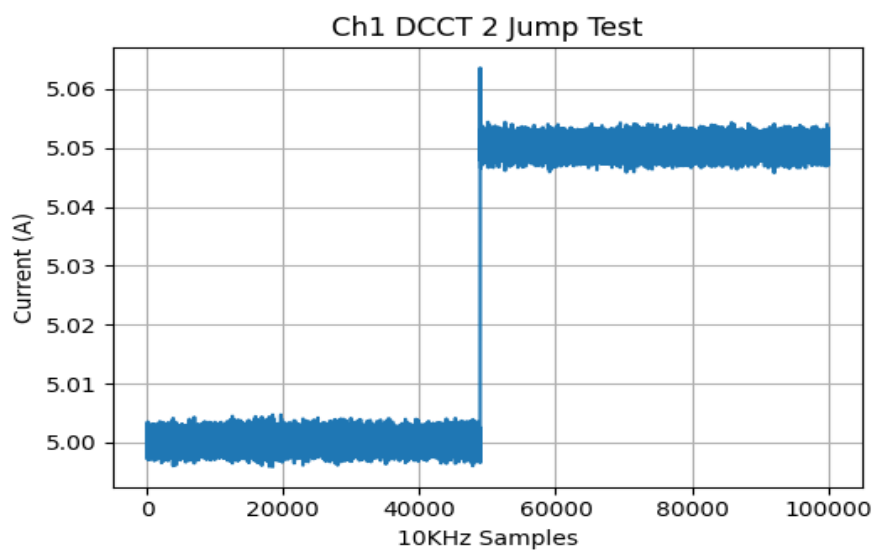
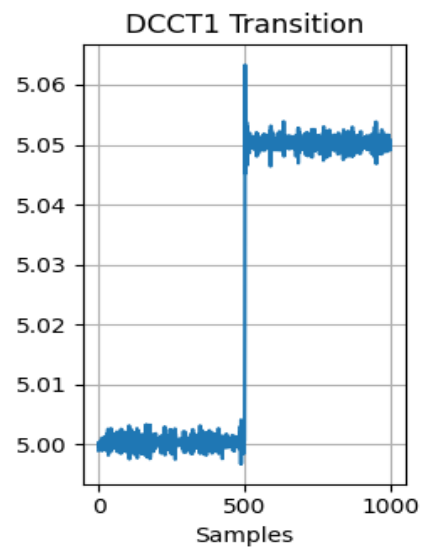
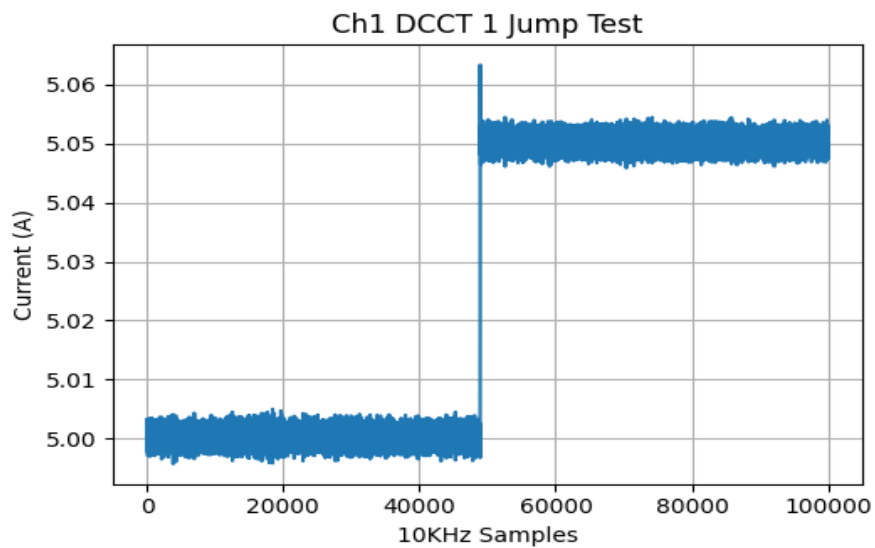
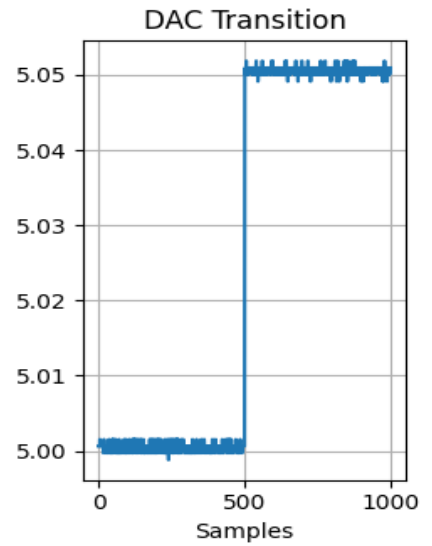
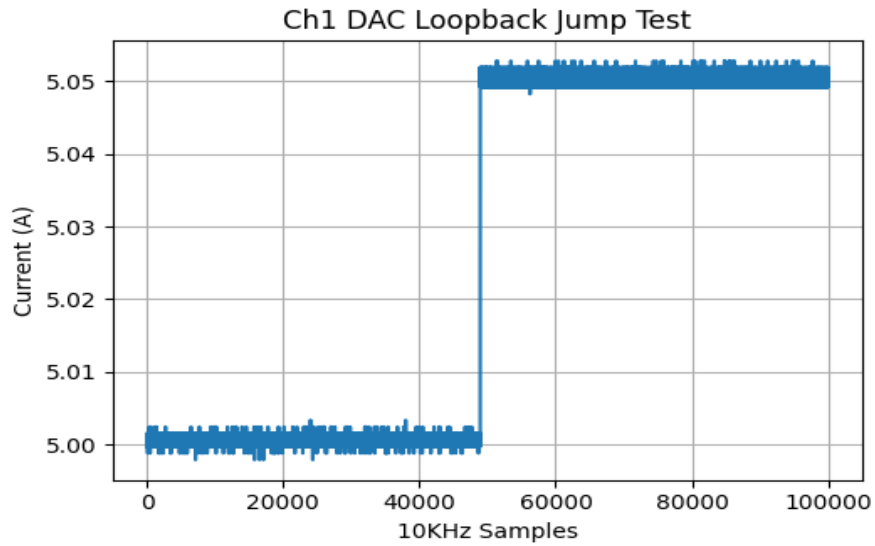
Channel 1

ATE Fault Tests for Channel 1	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

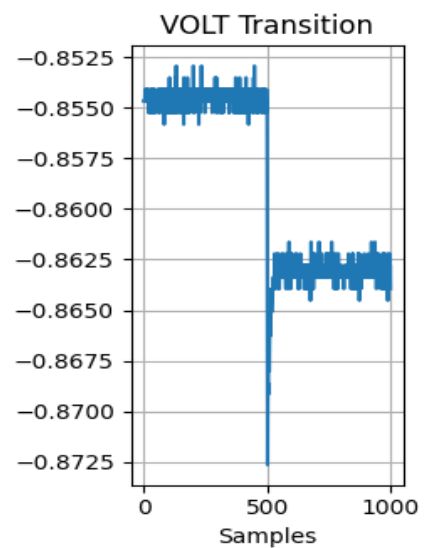
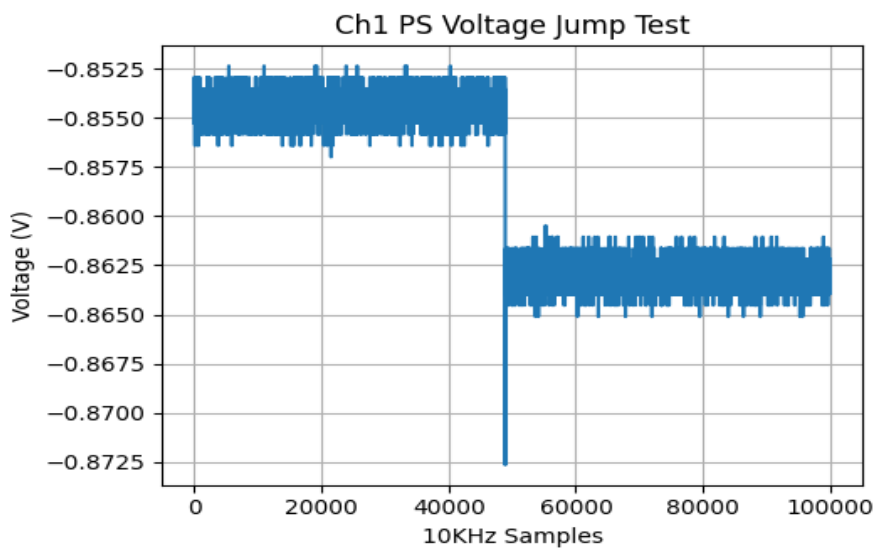
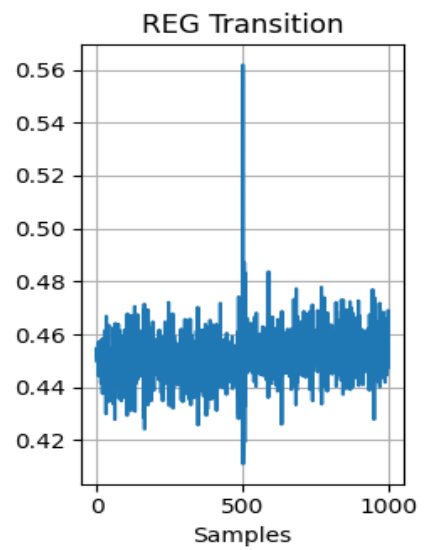
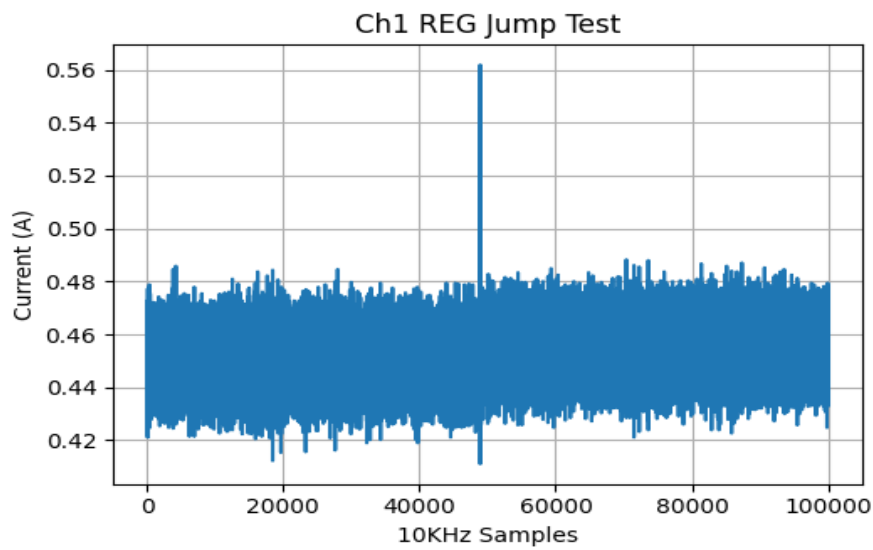
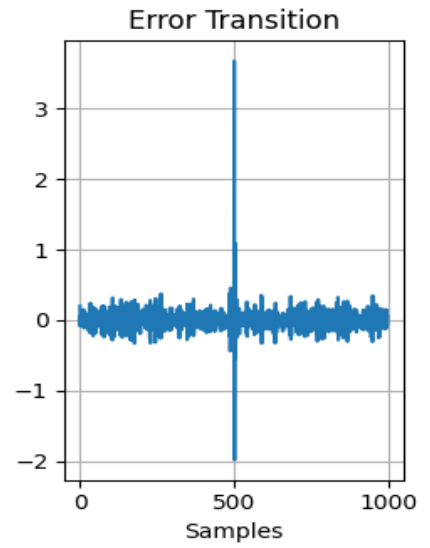
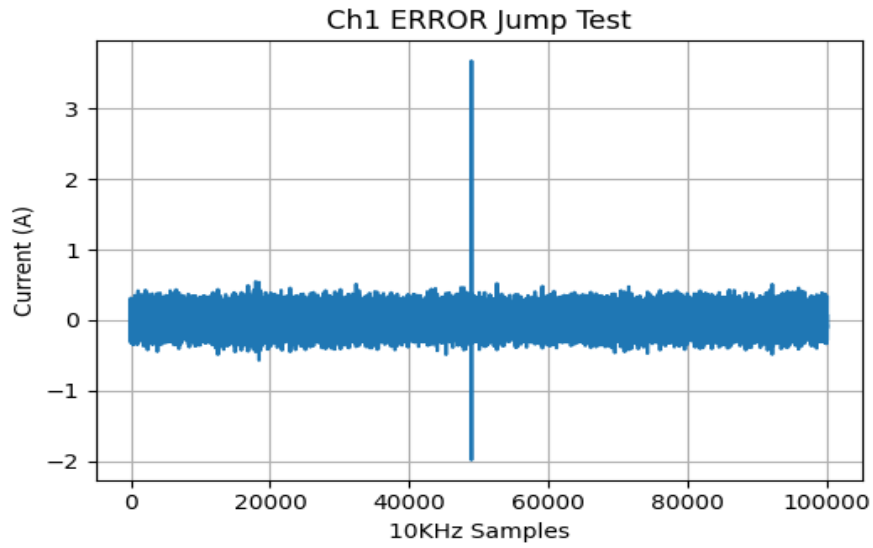
Power Supply Regulation for Channel 1:



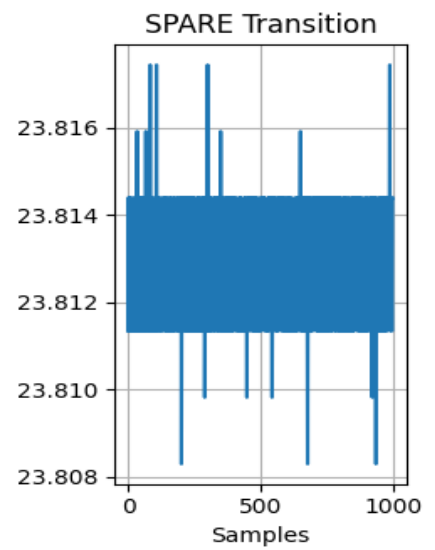
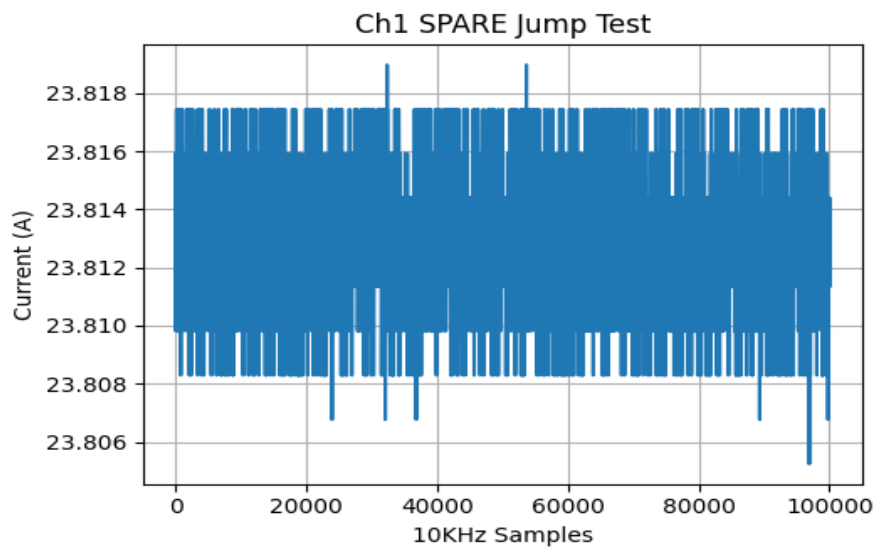
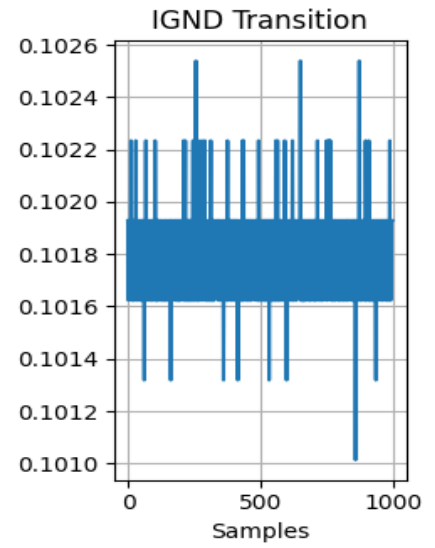
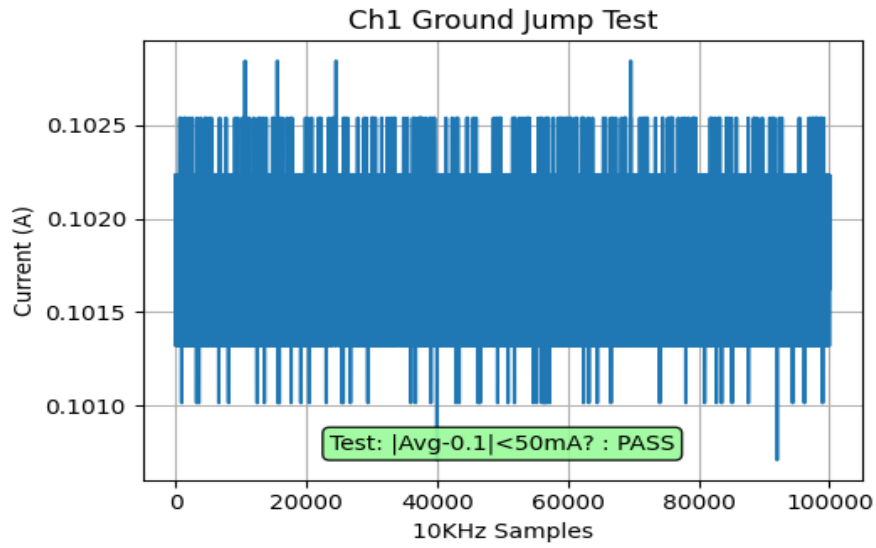
Jump Test Results: CH1



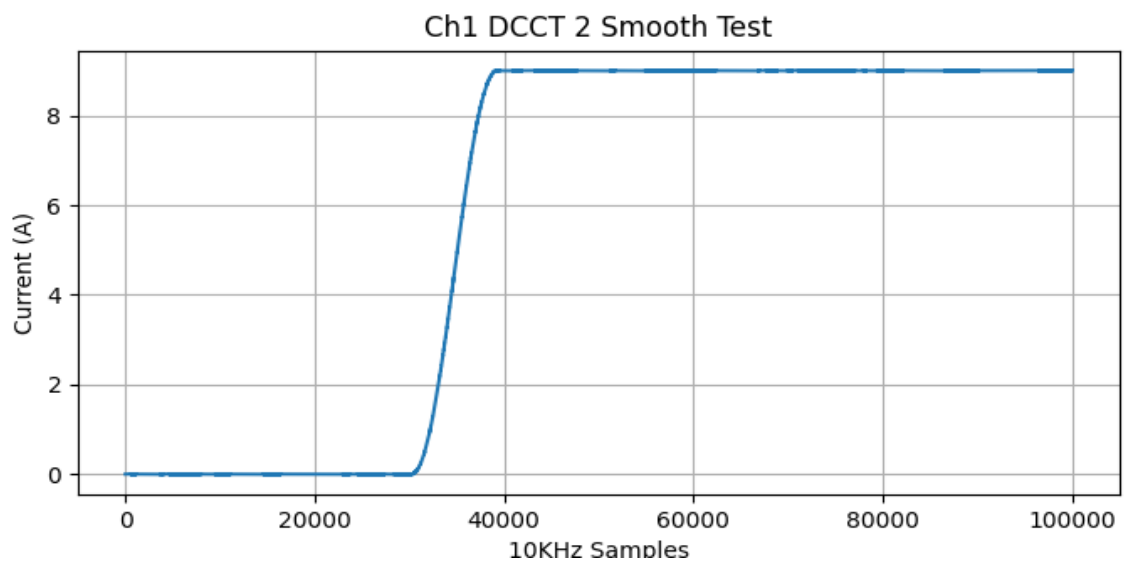
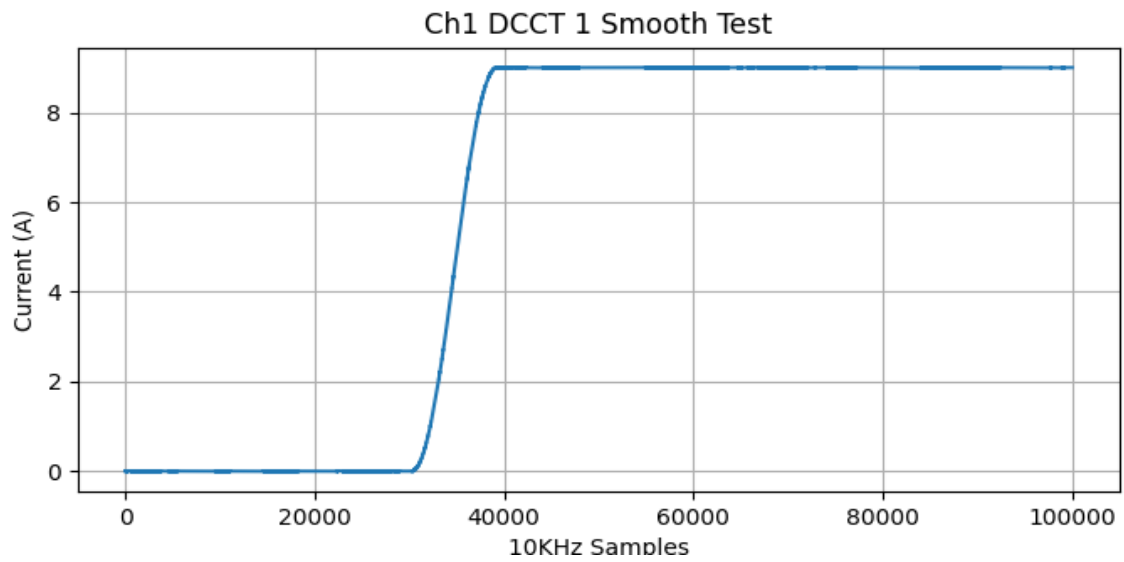
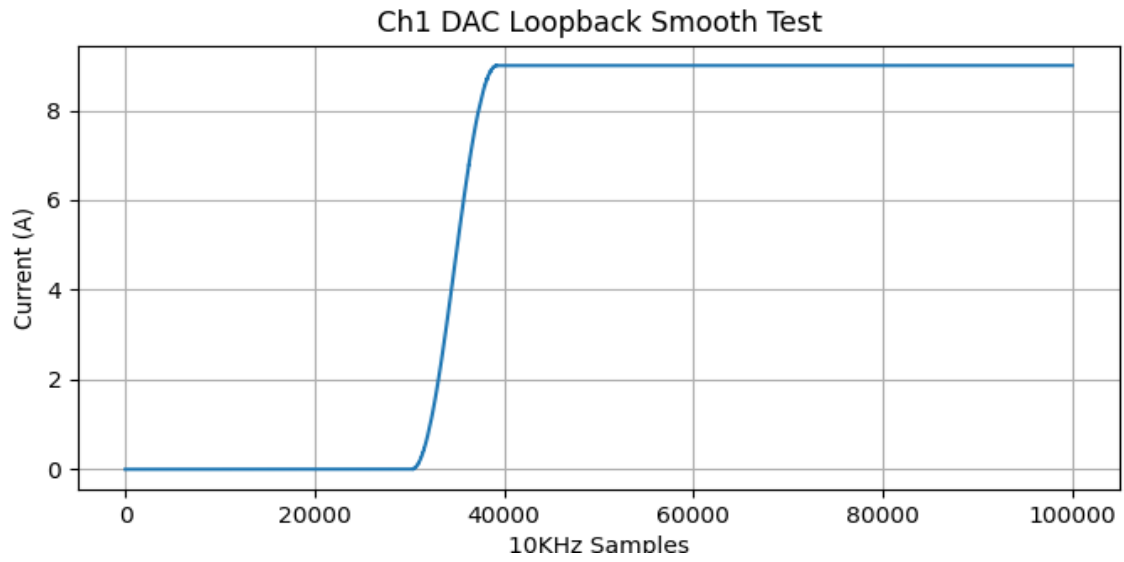
Jump Test Results: CH1



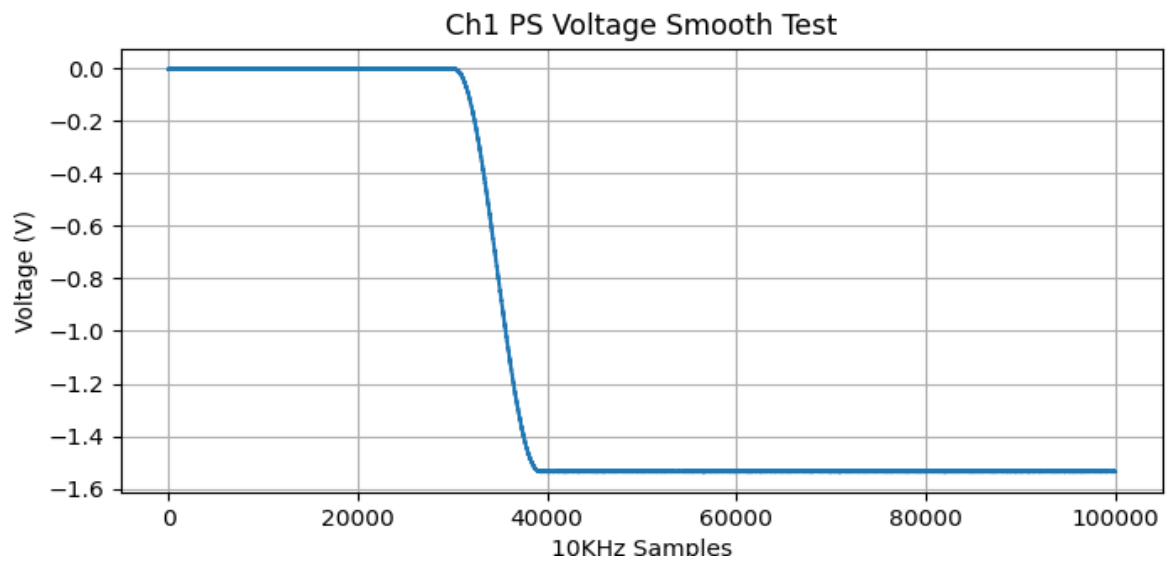
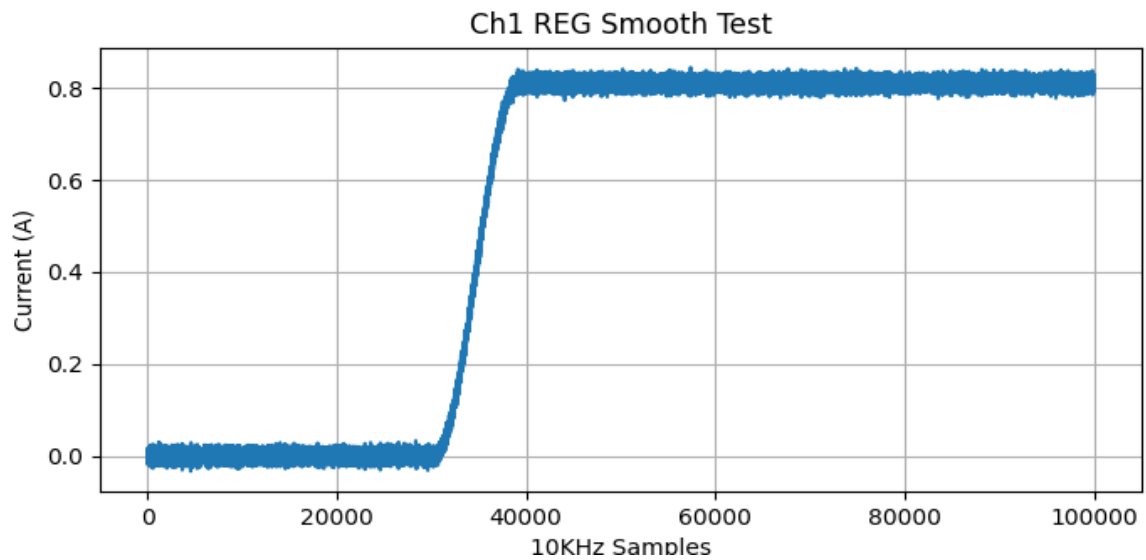
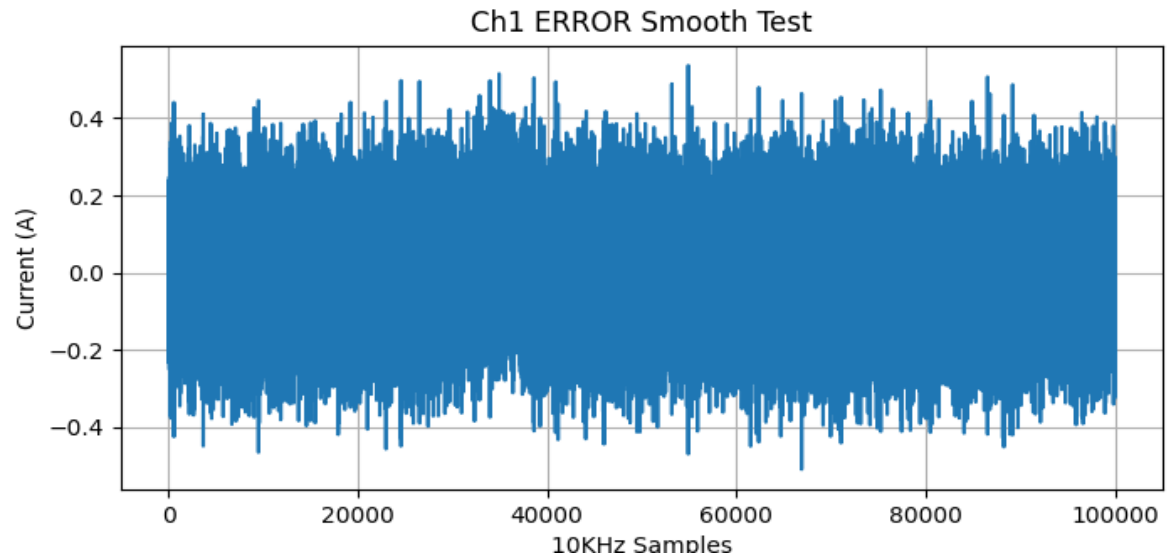
Jump Test Results: CH1



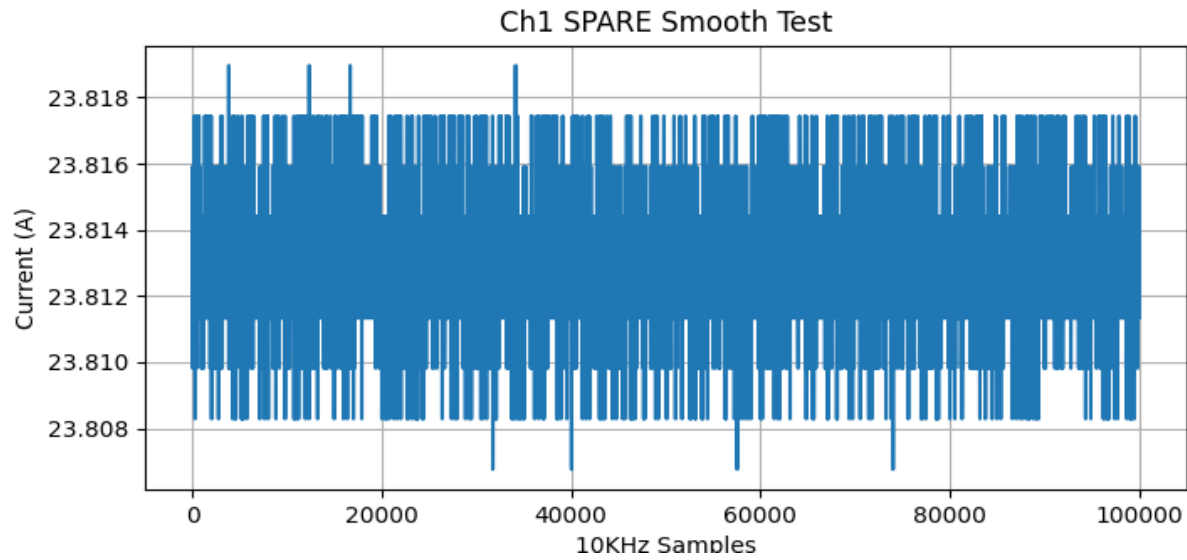
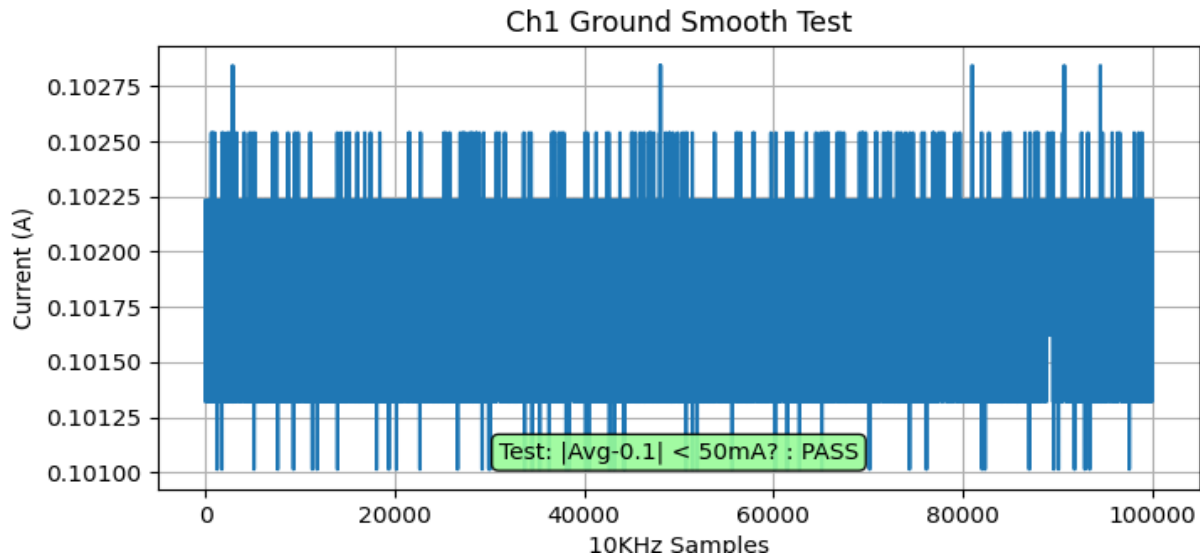
Smooth Test Results: CH1



Smooth Test Results: CH1



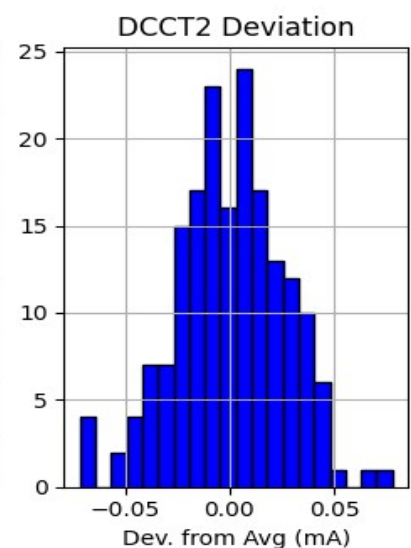
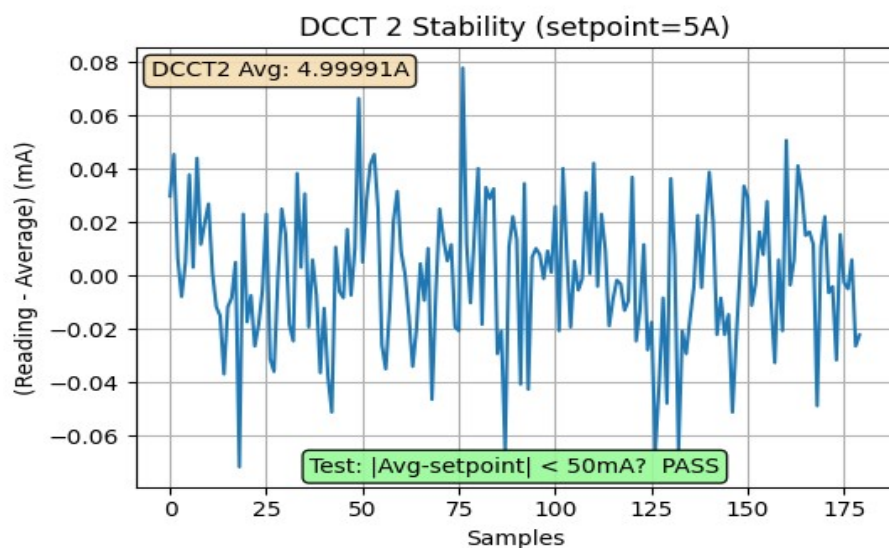
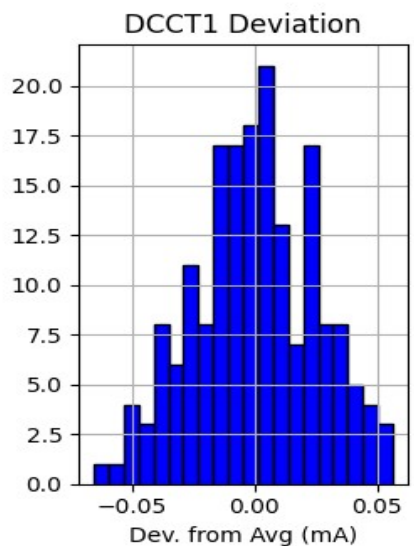
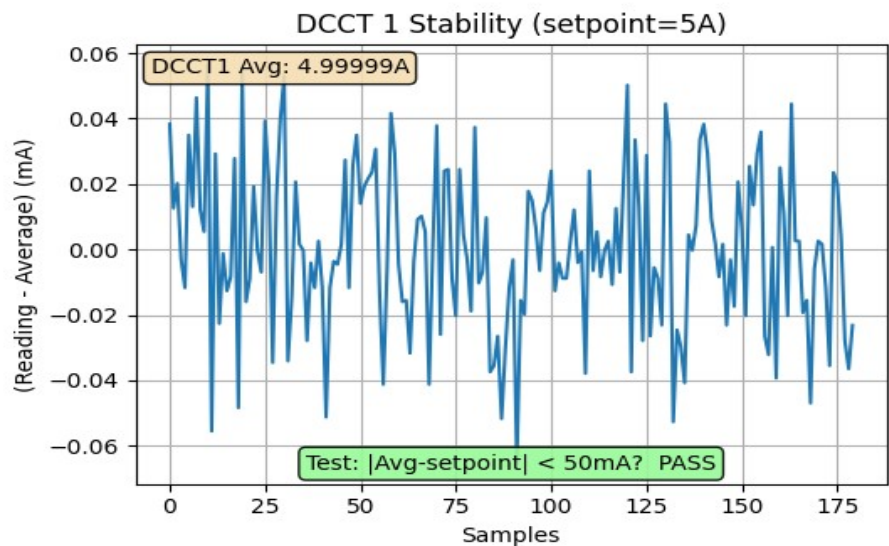
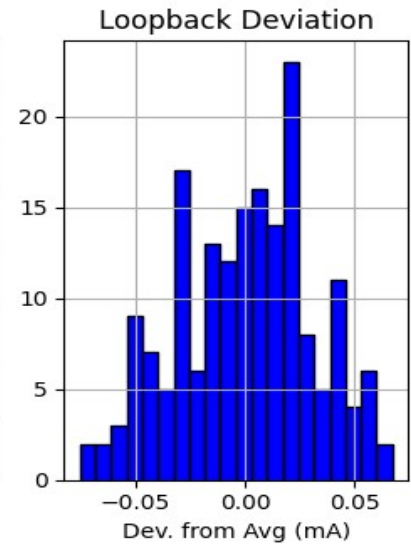
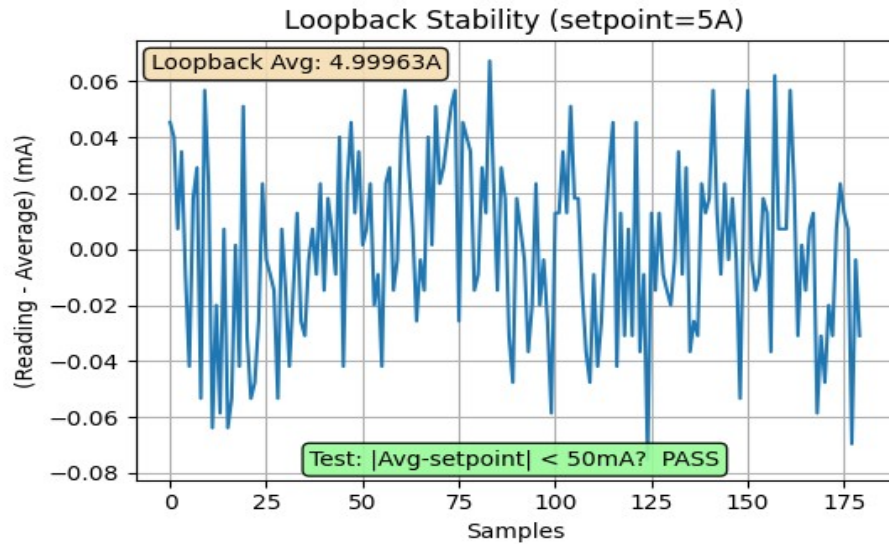
Smooth Test Results: CH1



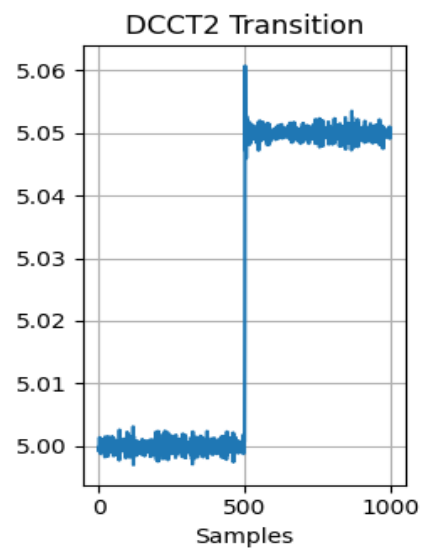
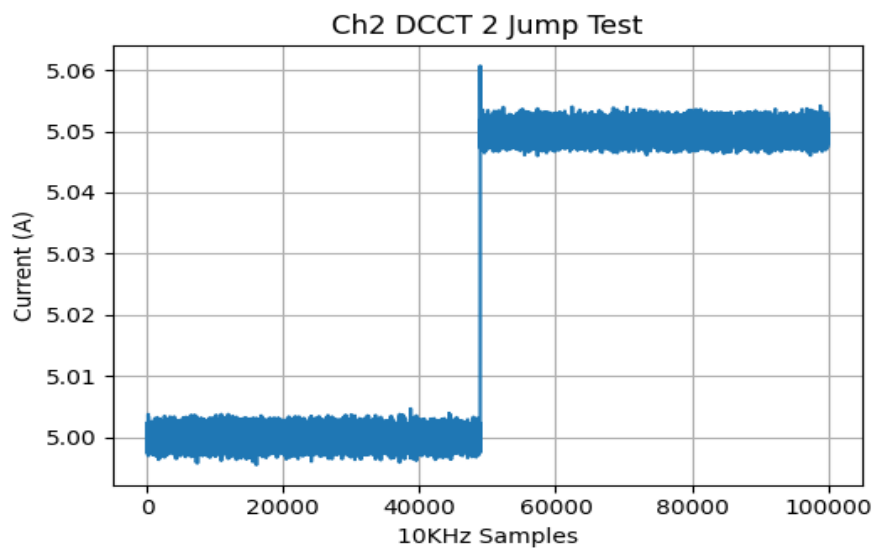
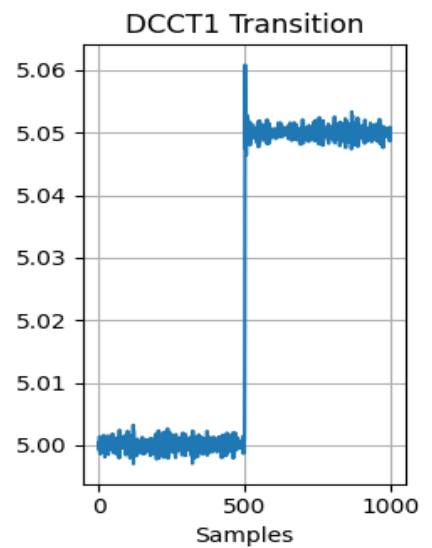
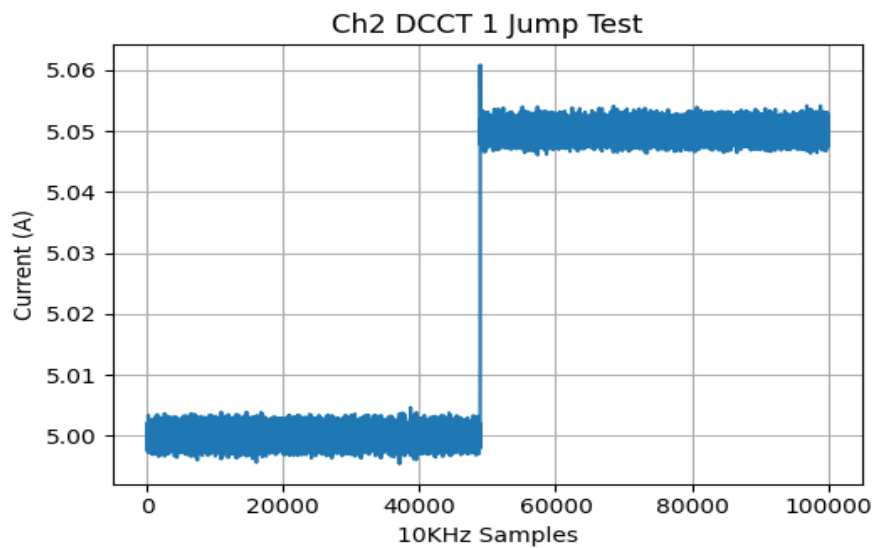
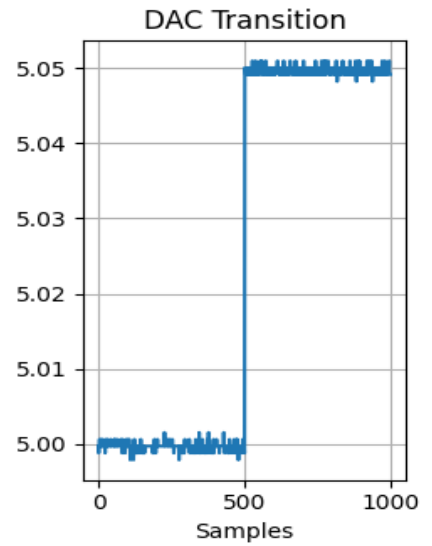
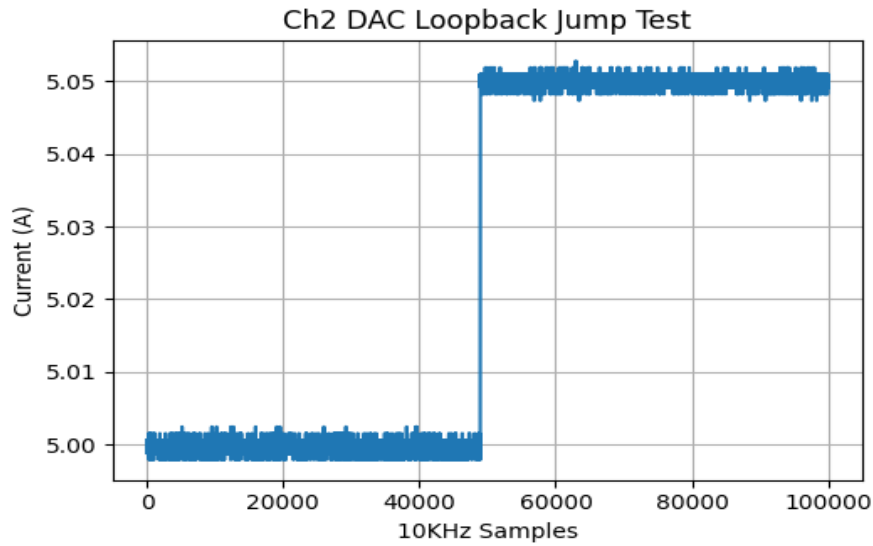
Channel 2

ATE Fault Tests for Channel 2	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

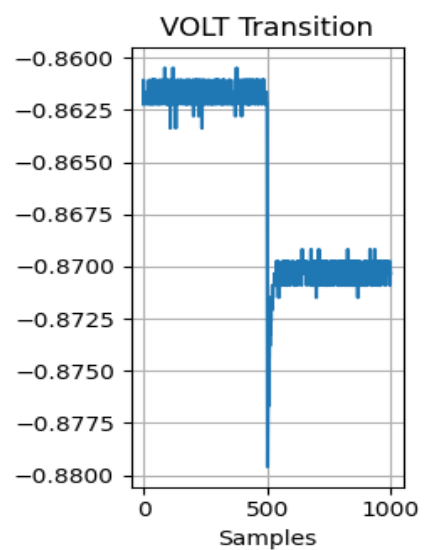
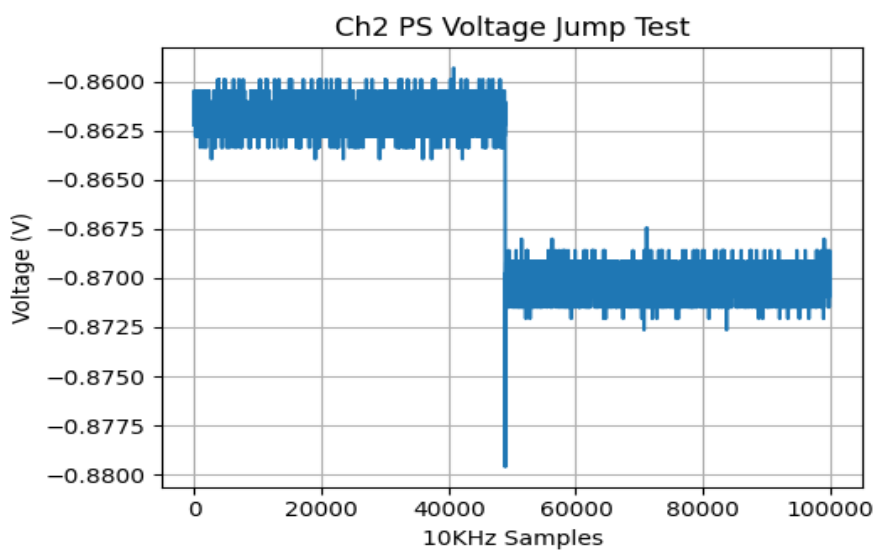
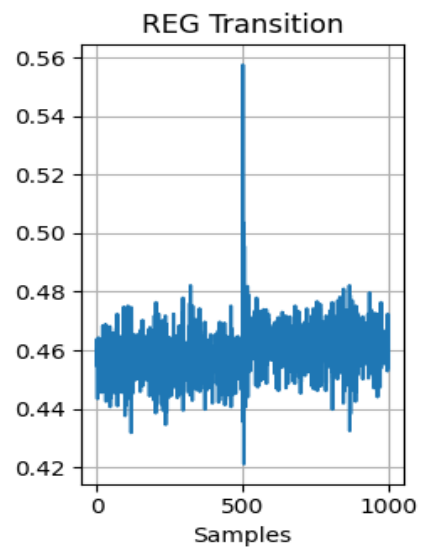
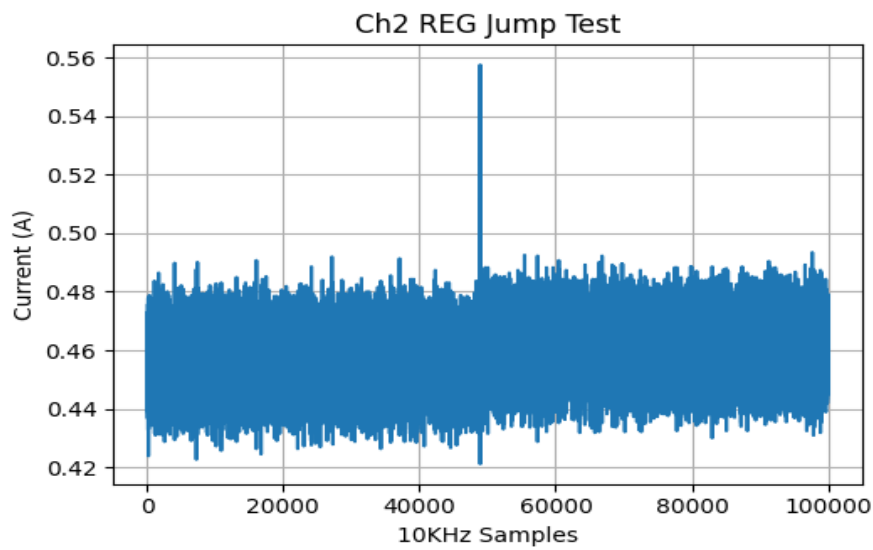
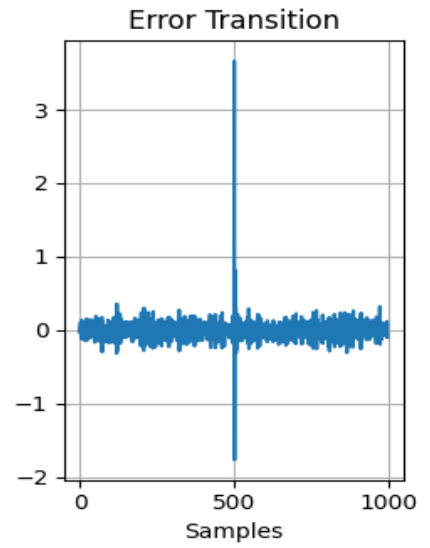
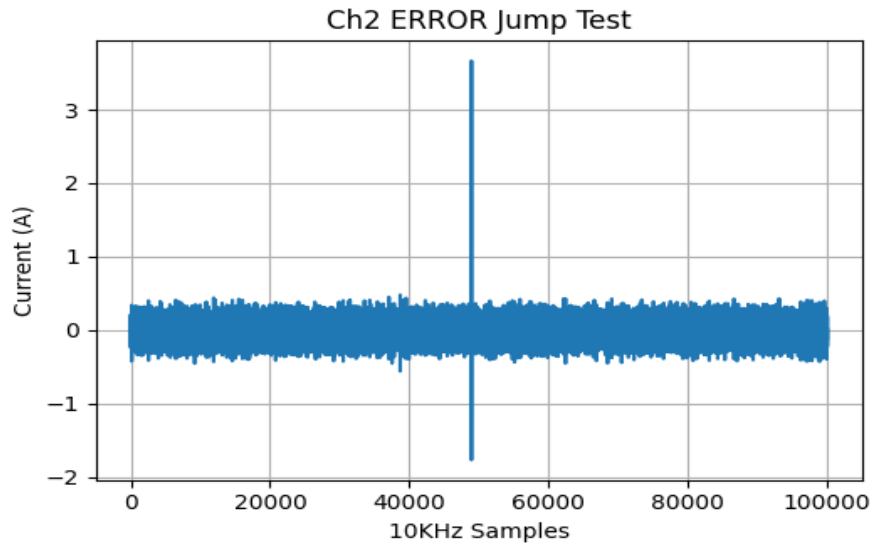
Power Supply Regulation for Channel 2:



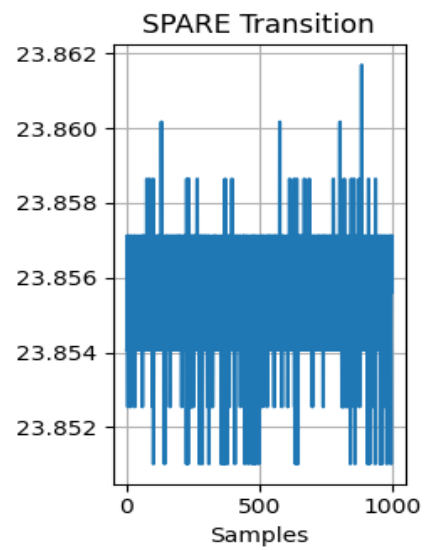
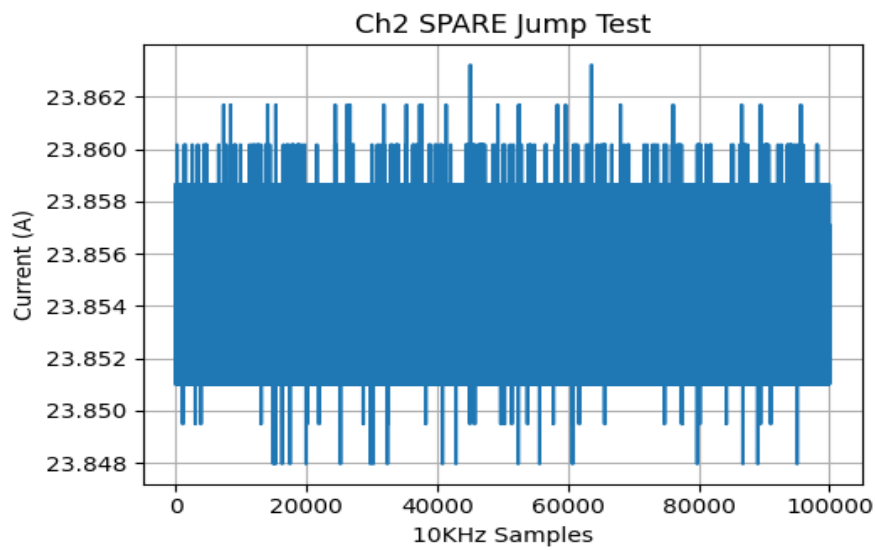
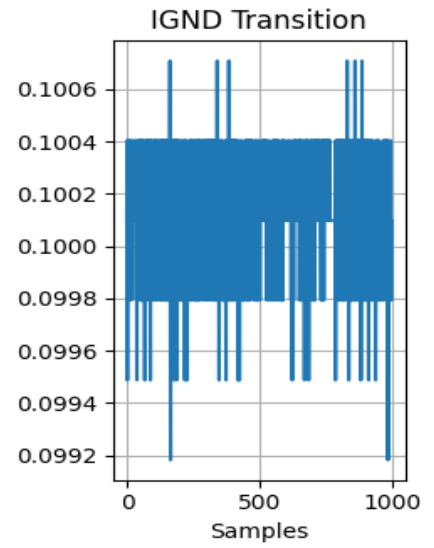
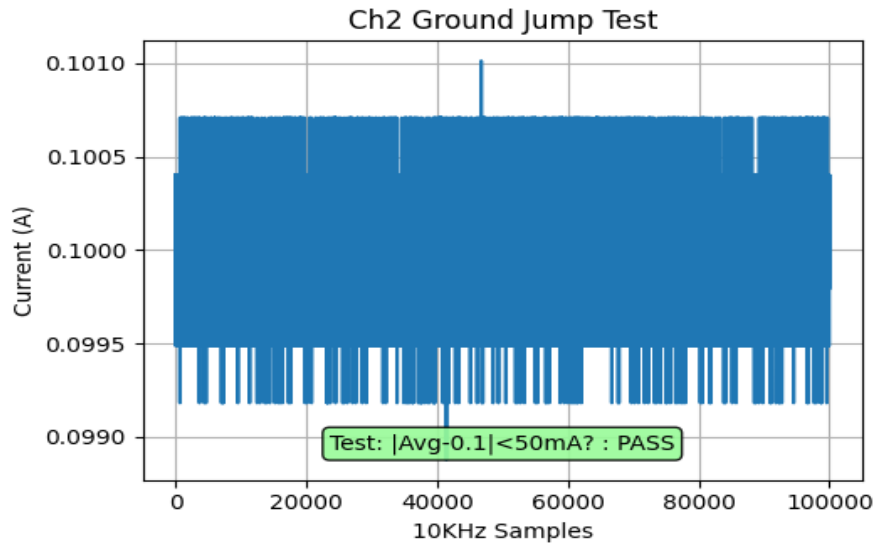
Jump Test Results: CH2



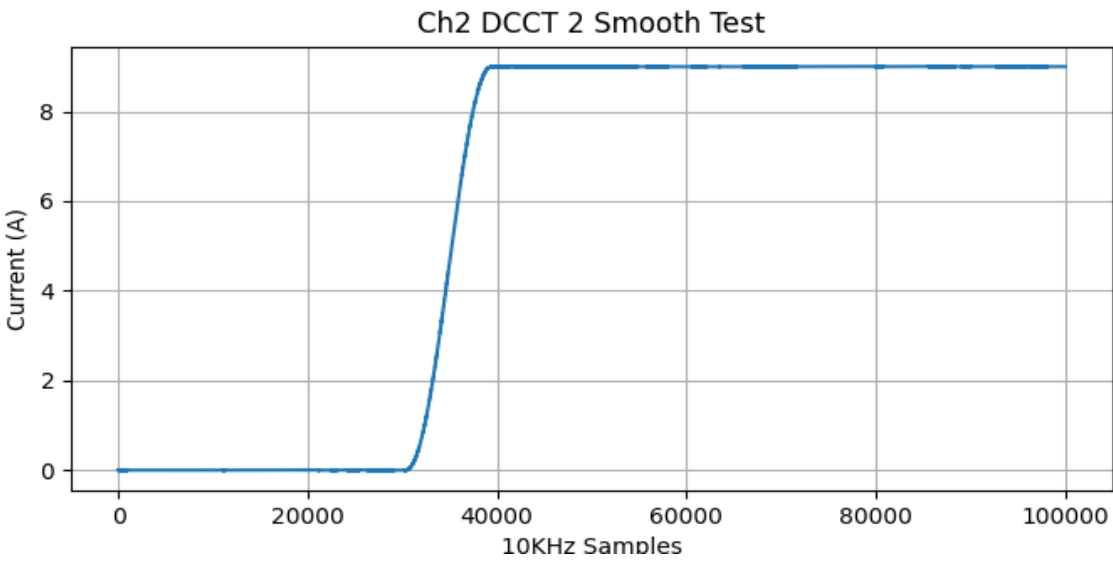
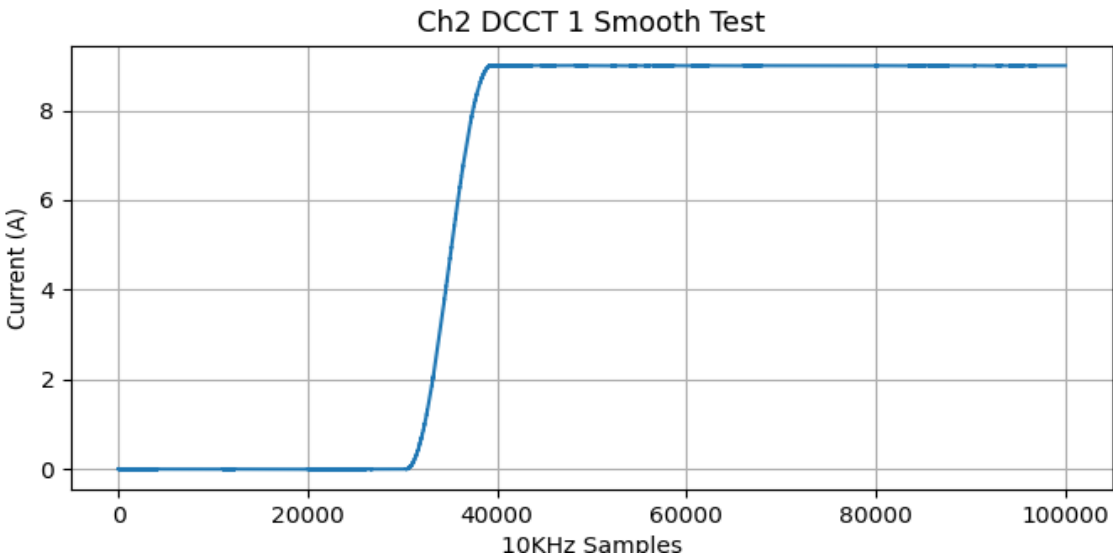
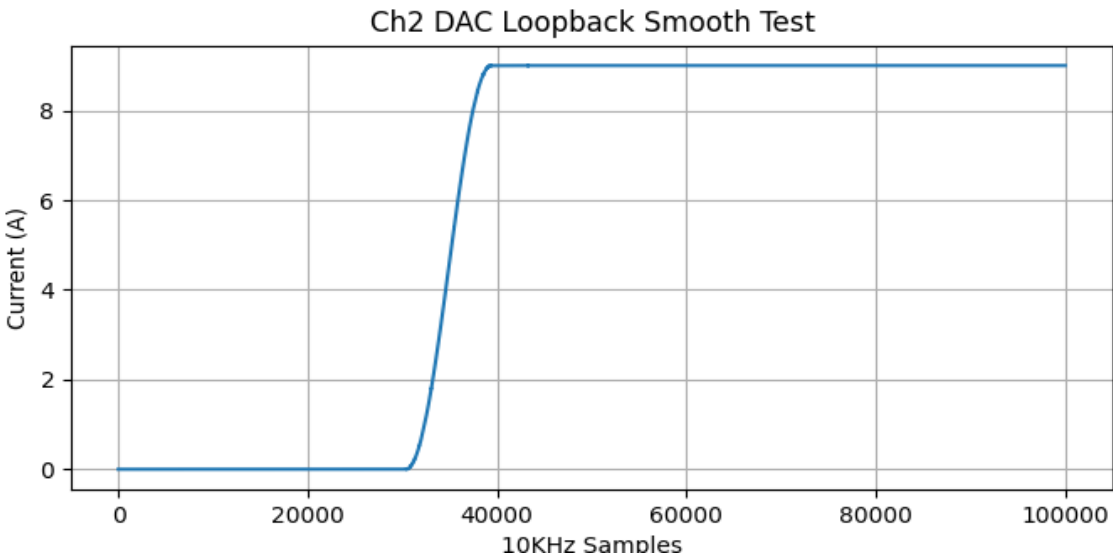
Jump Test Results: CH2



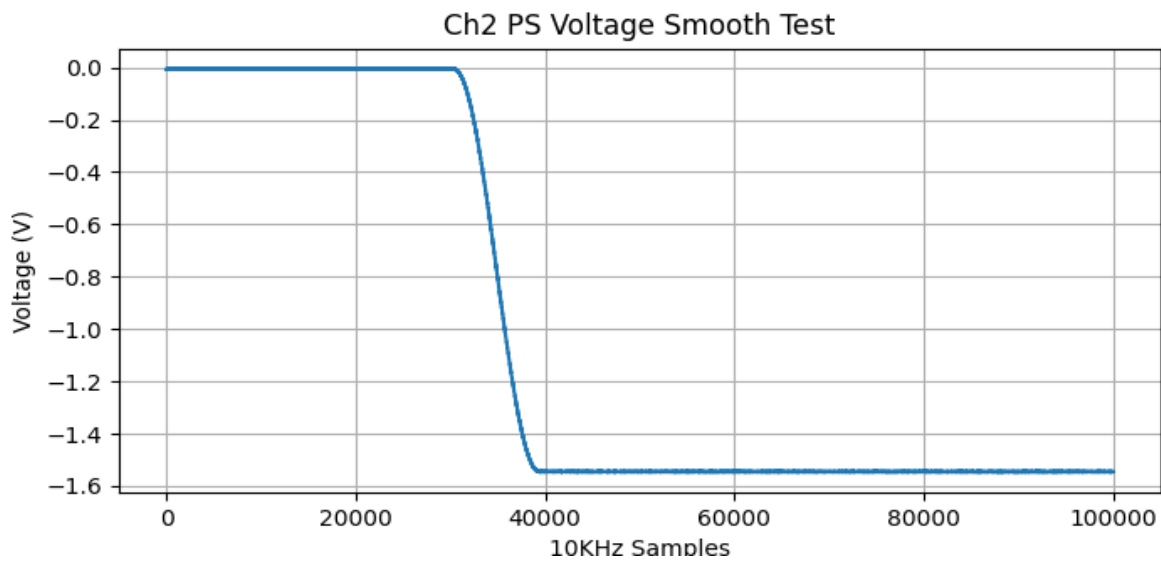
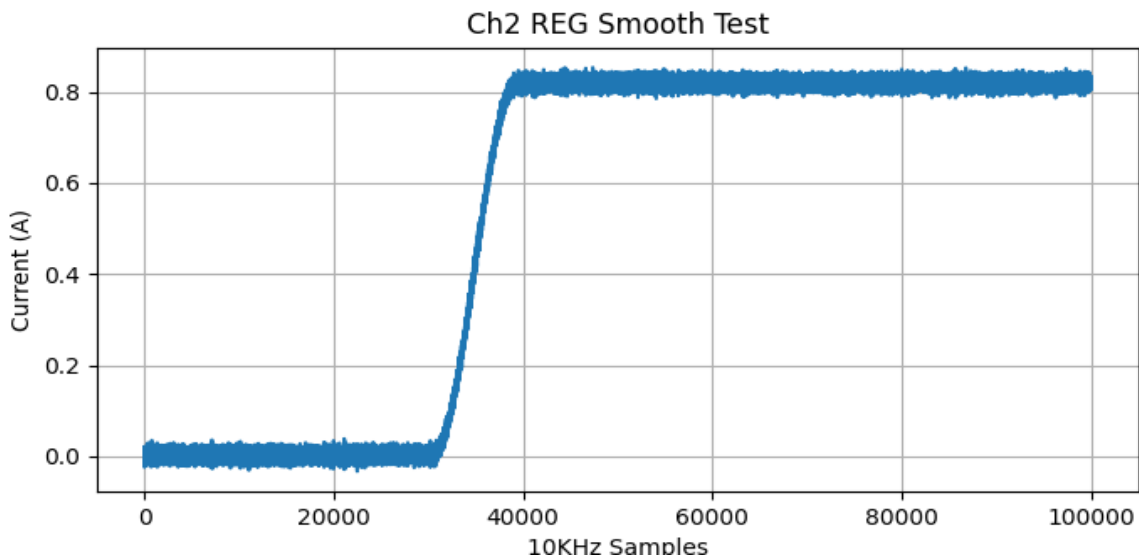
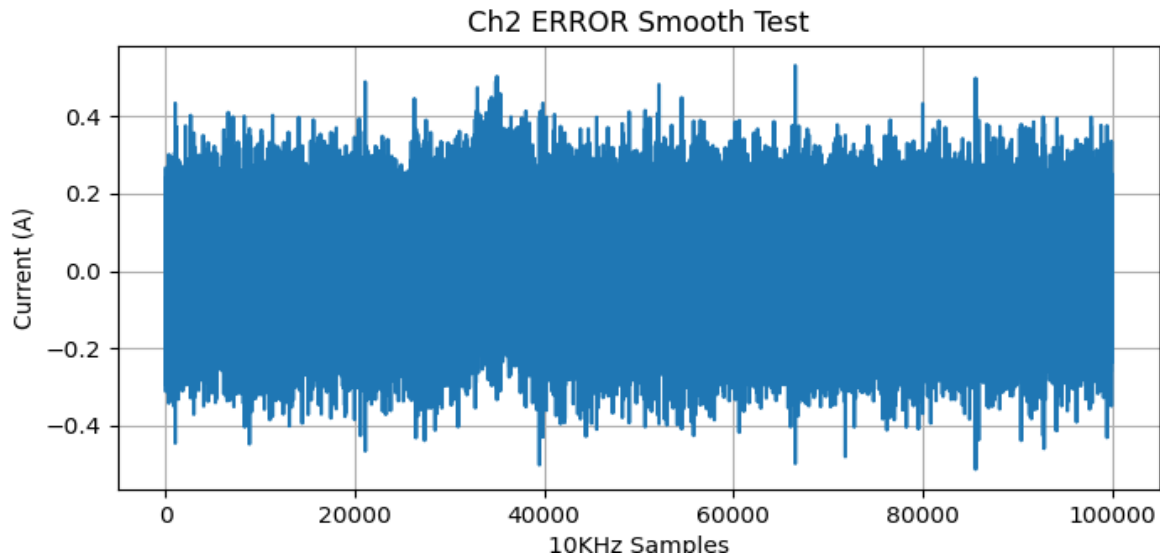
Jump Test Results: CH2



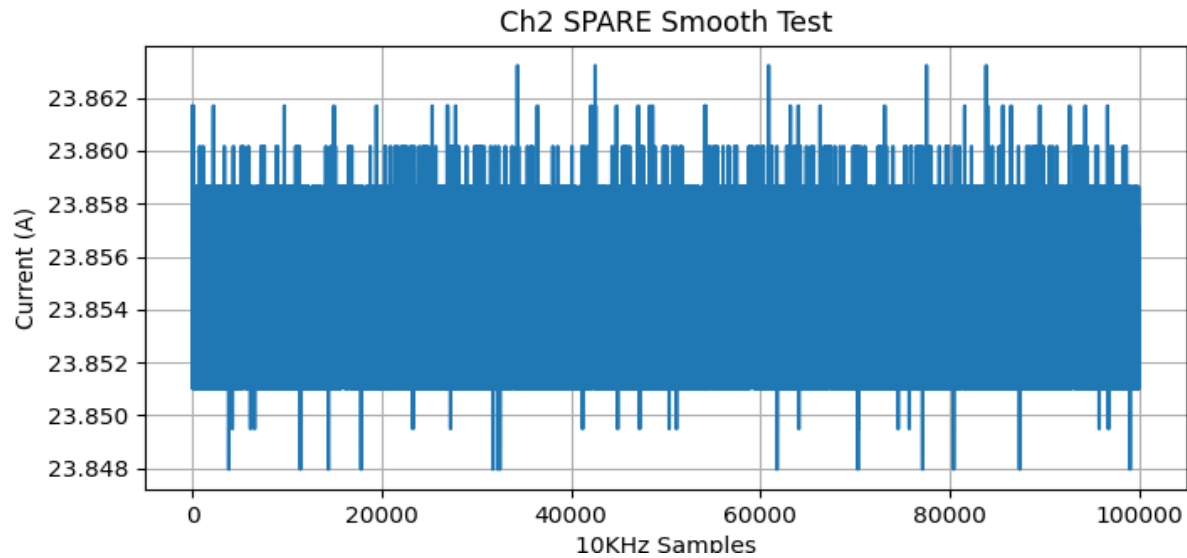
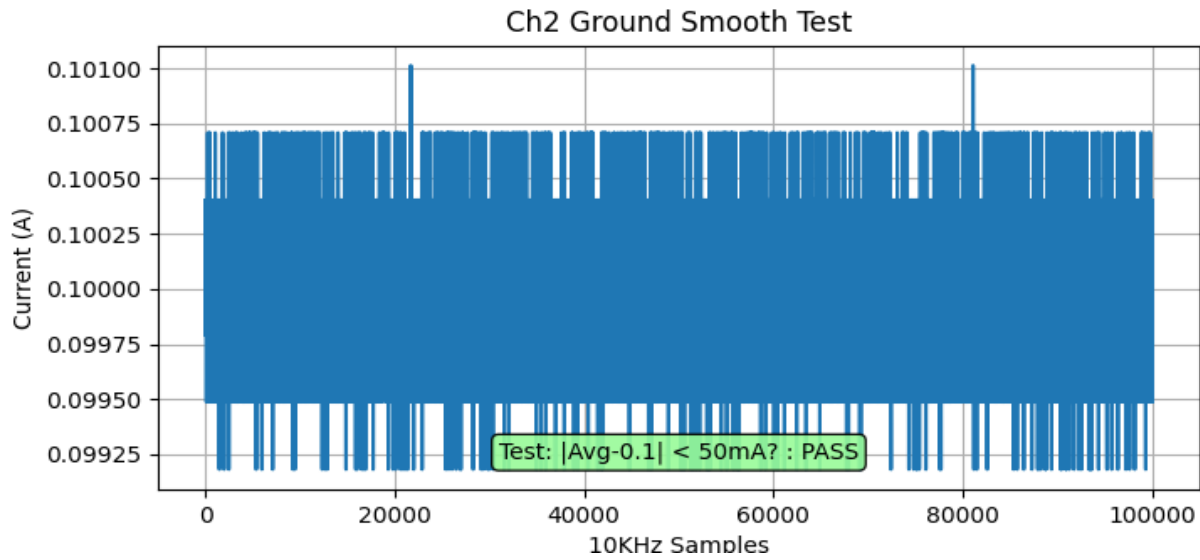
Smooth Test Results: CH2



Smooth Test Results: CH2



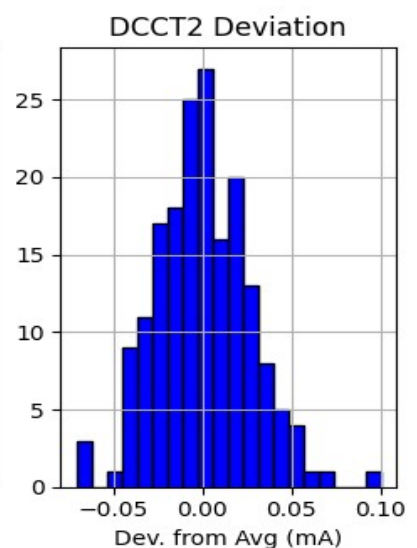
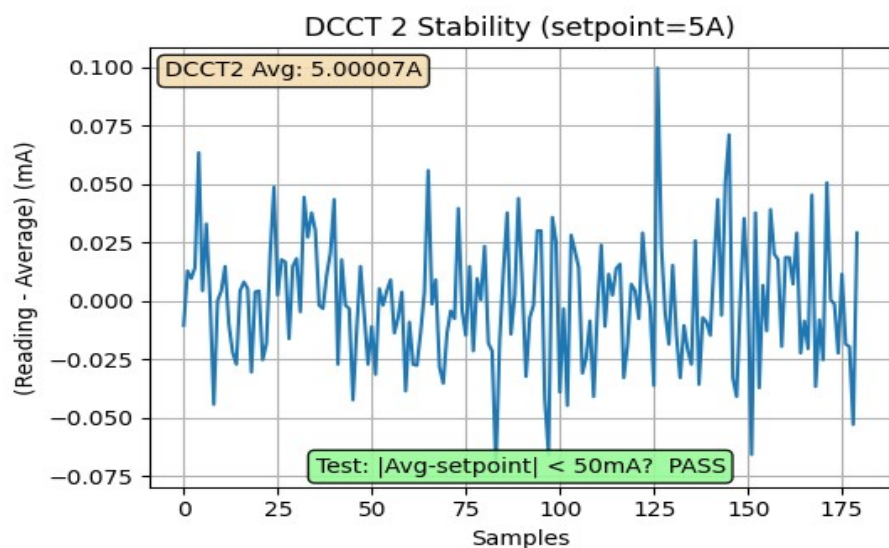
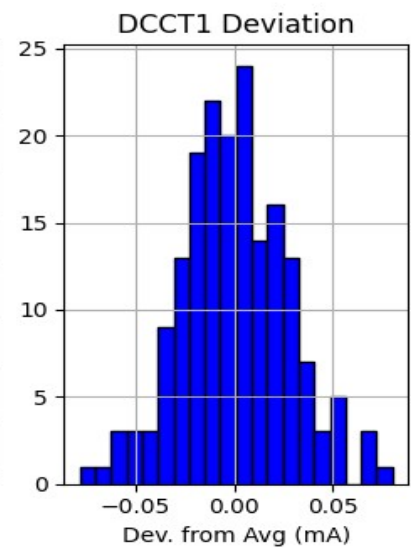
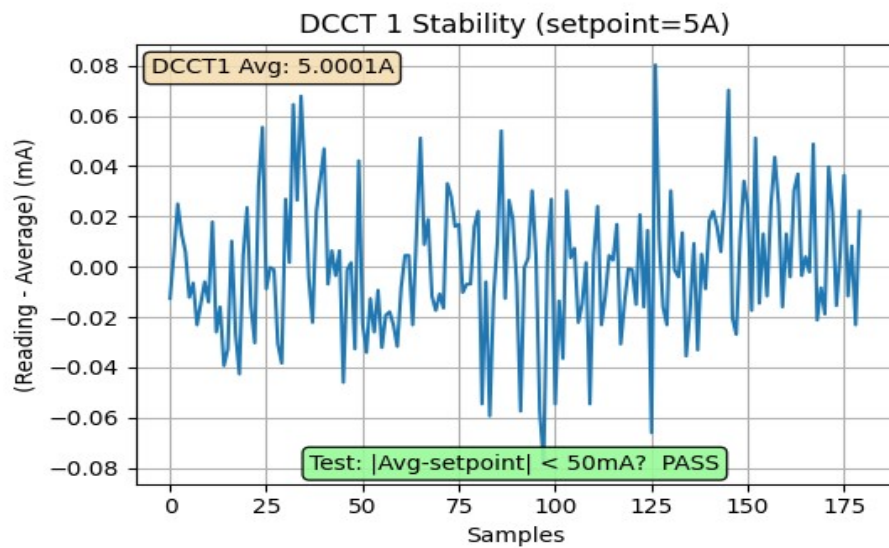
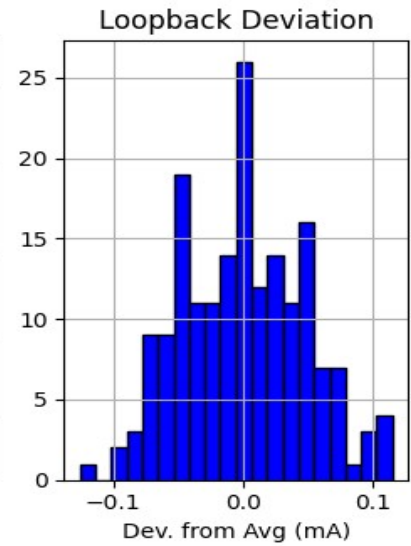
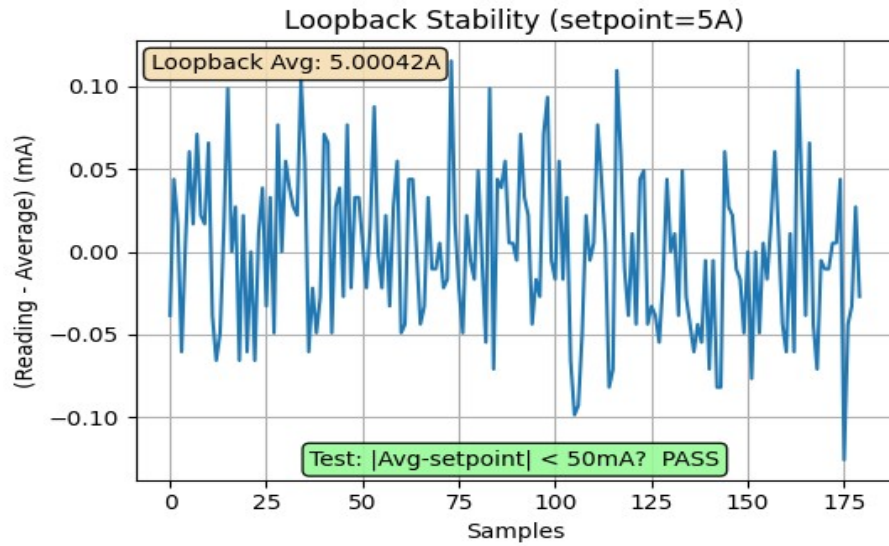
Smooth Test Results: CH2



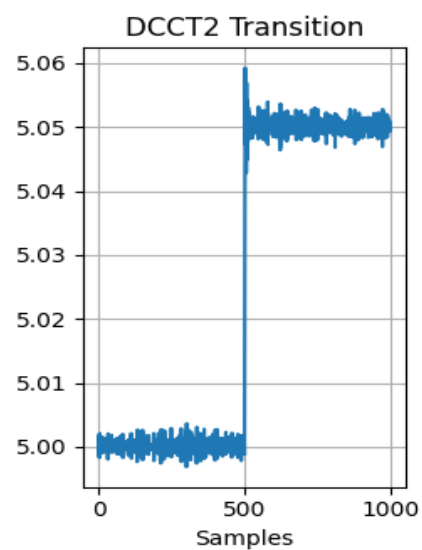
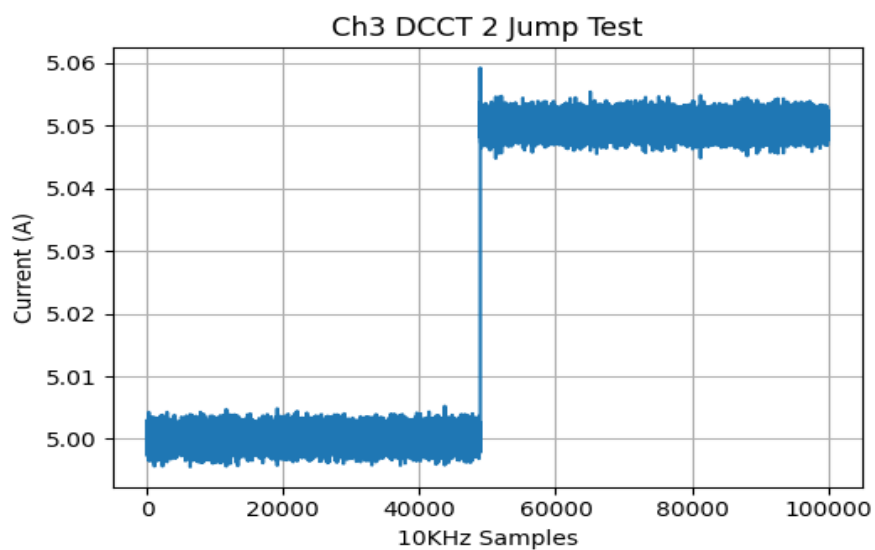
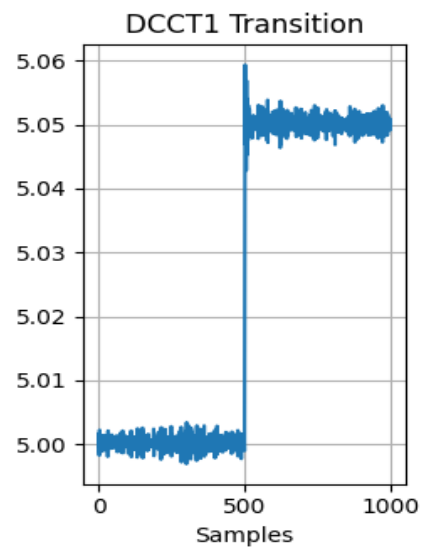
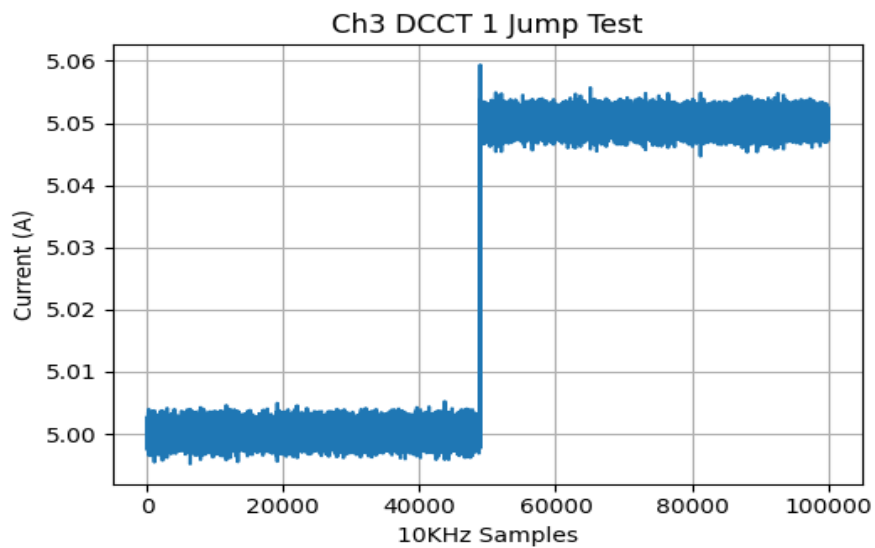
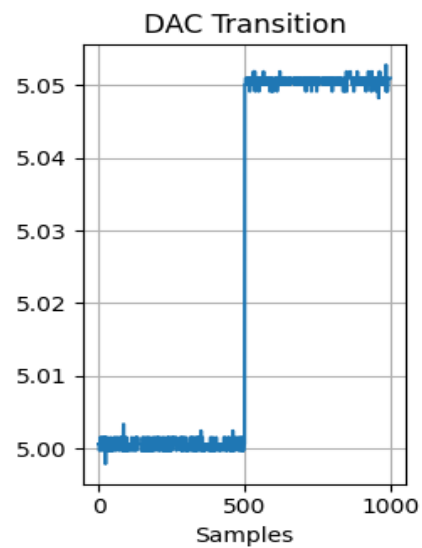
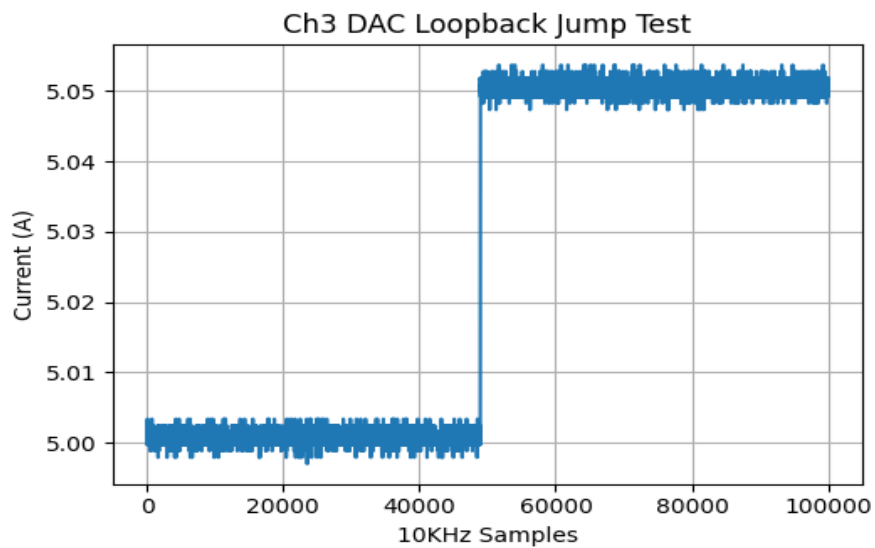
Channel 3

ATE Fault Tests for Channel 3	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

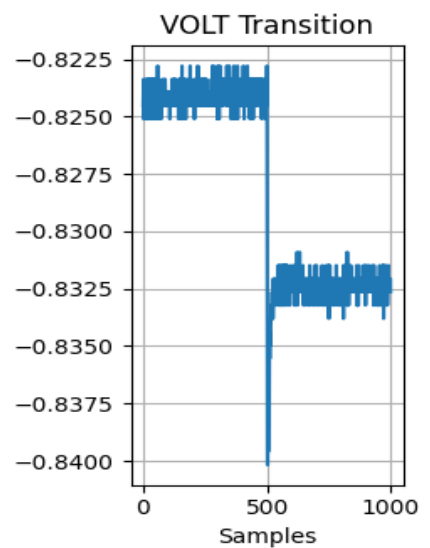
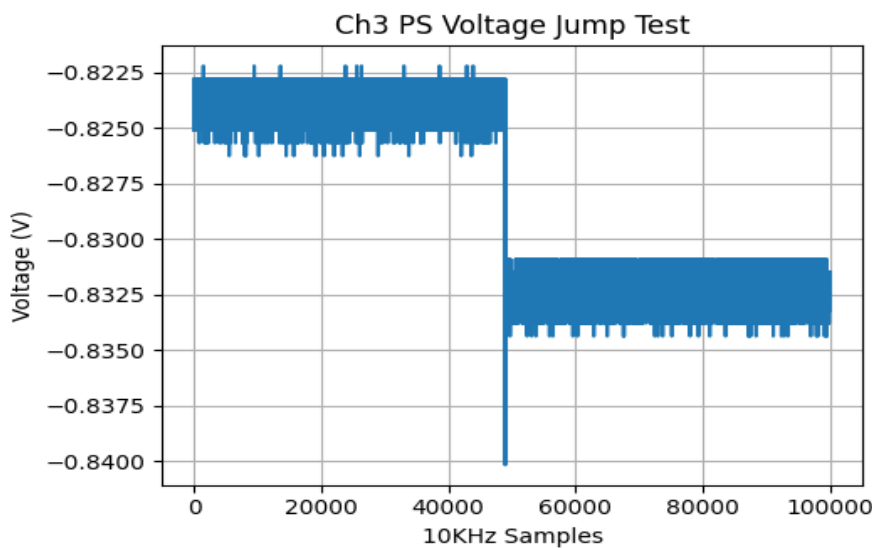
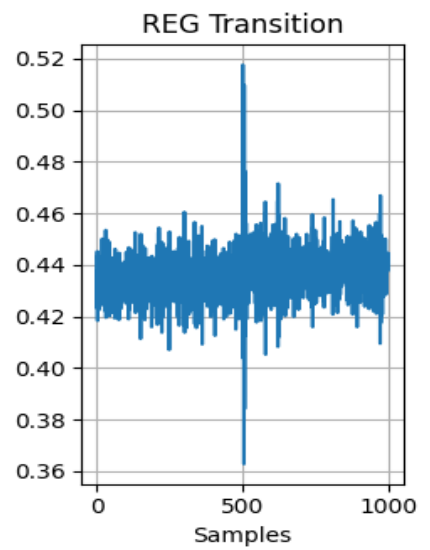
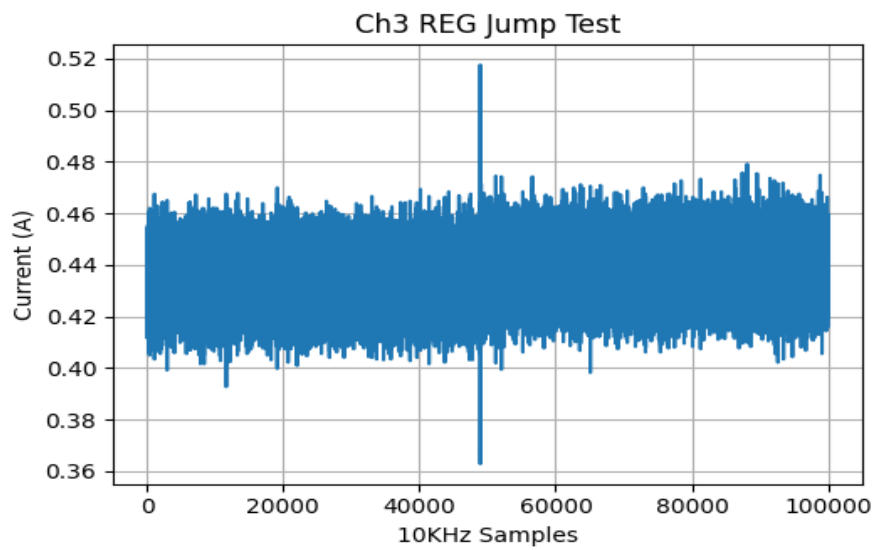
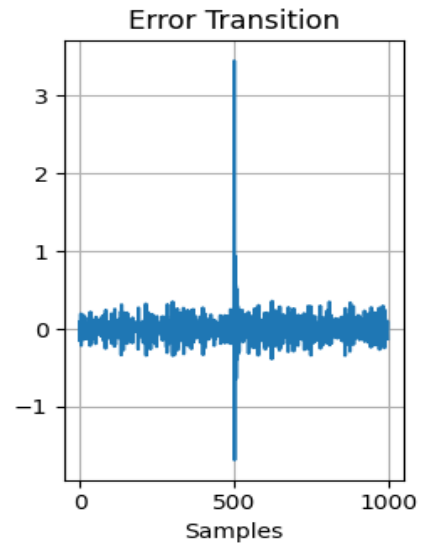
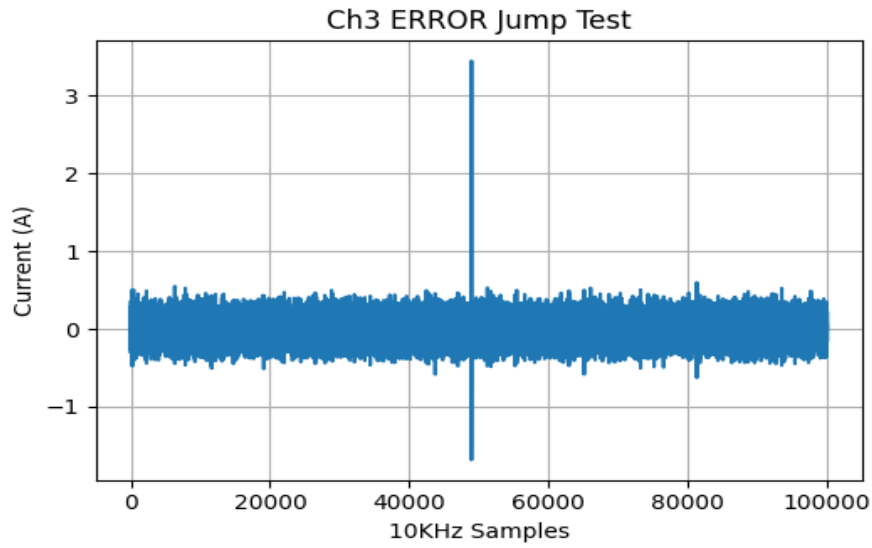
Power Supply Regulation for Channel 3:



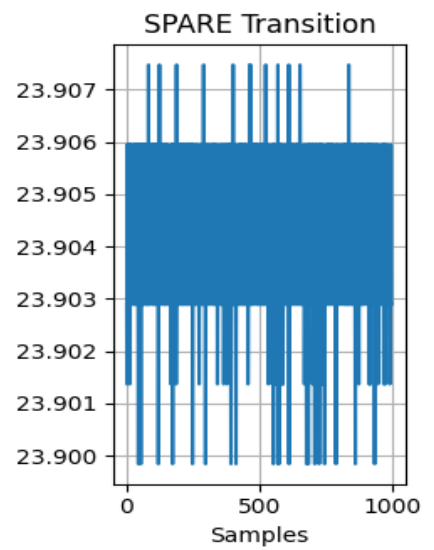
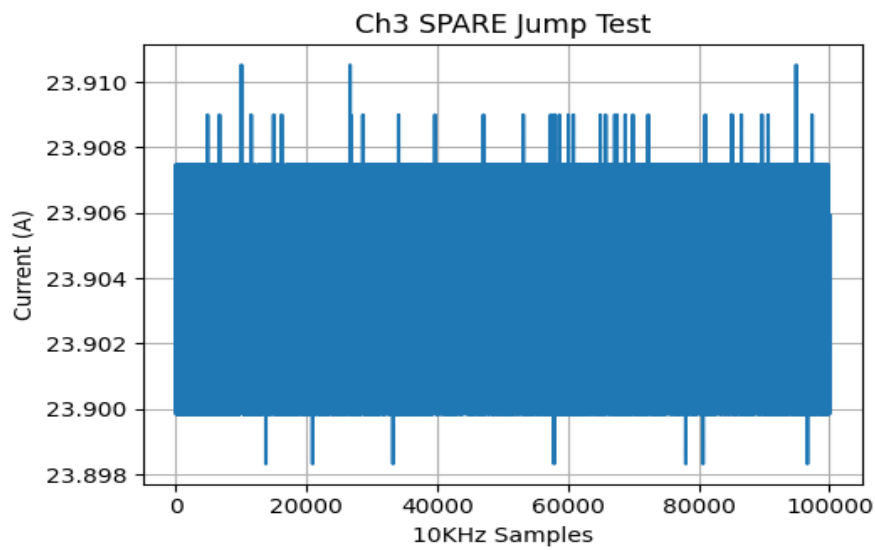
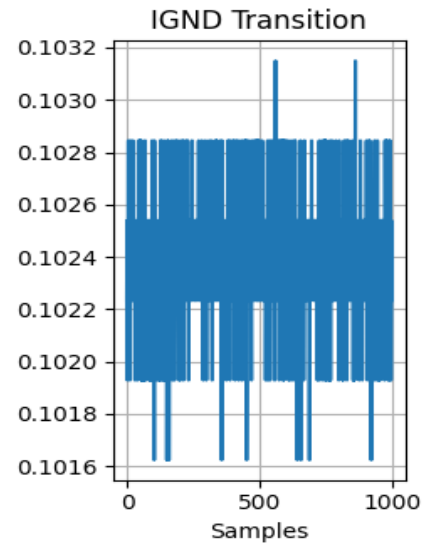
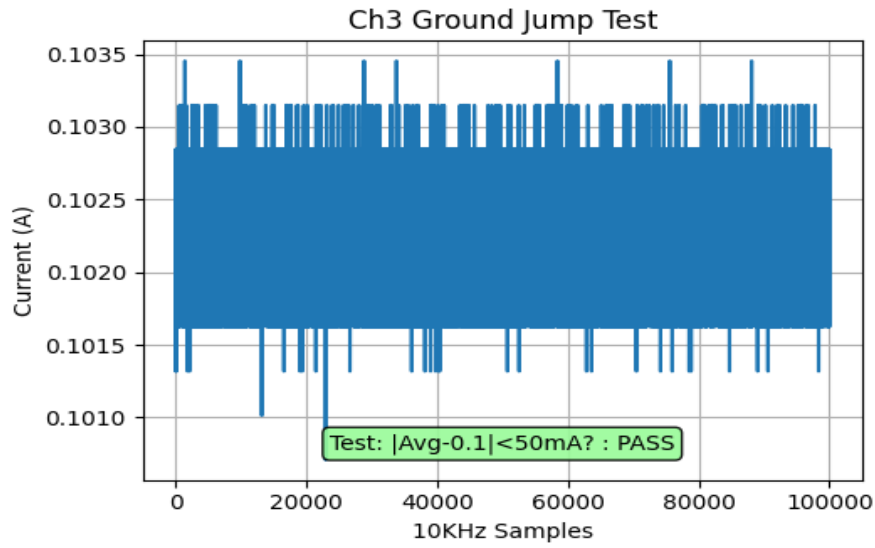
Jump Test Results: CH3



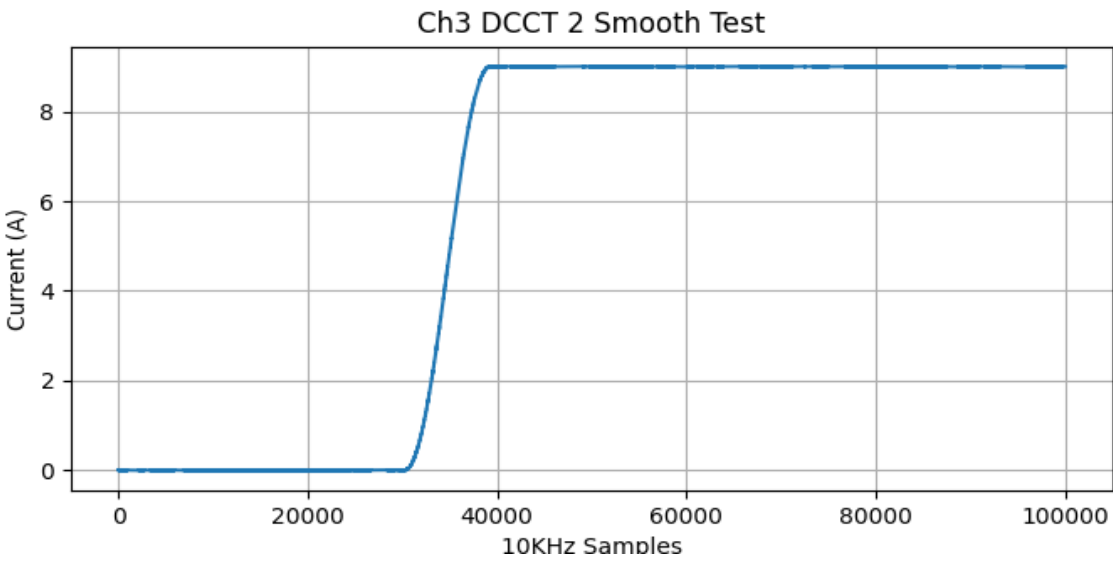
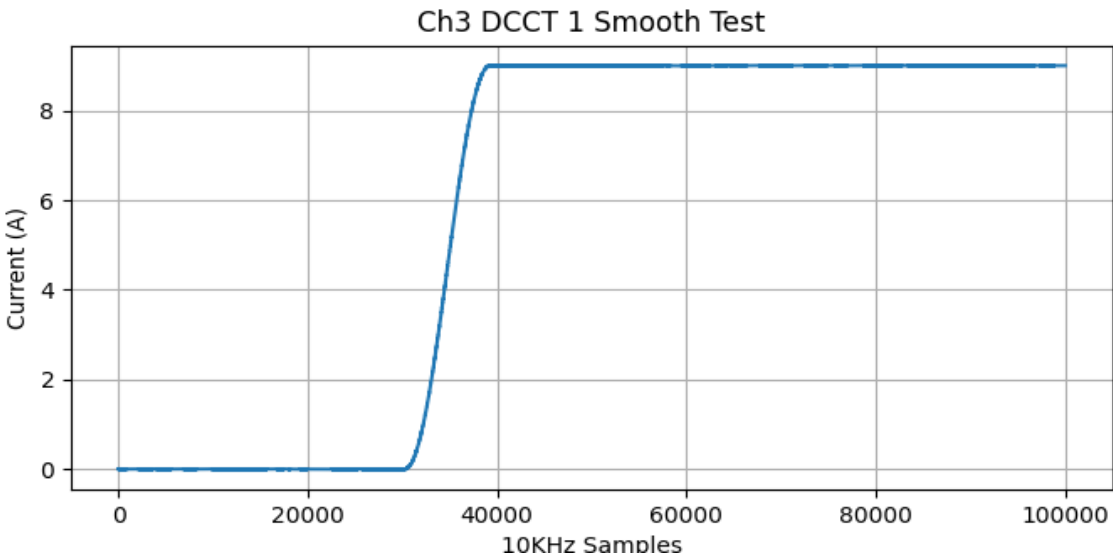
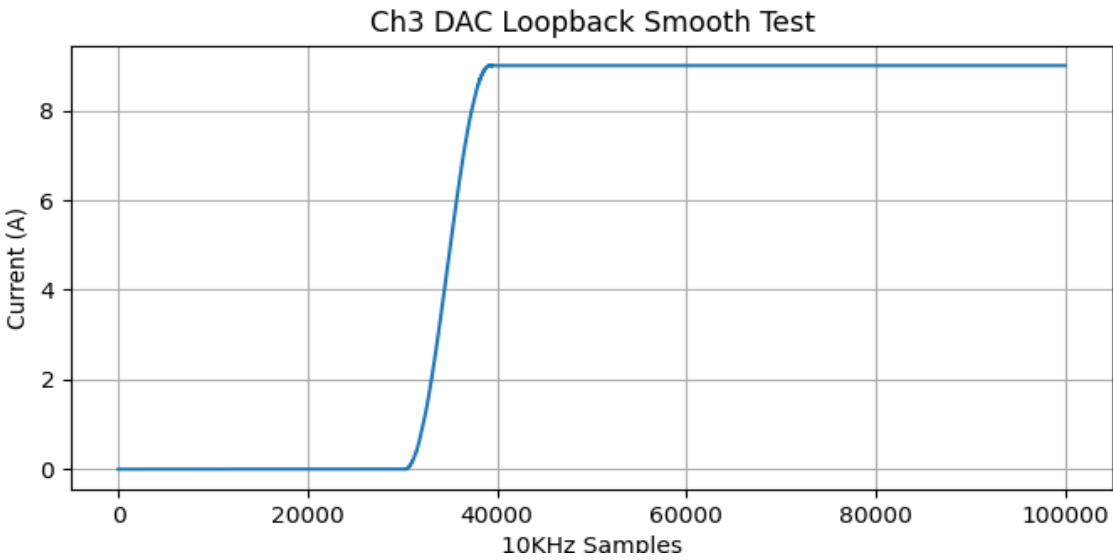
Jump Test Results: CH3



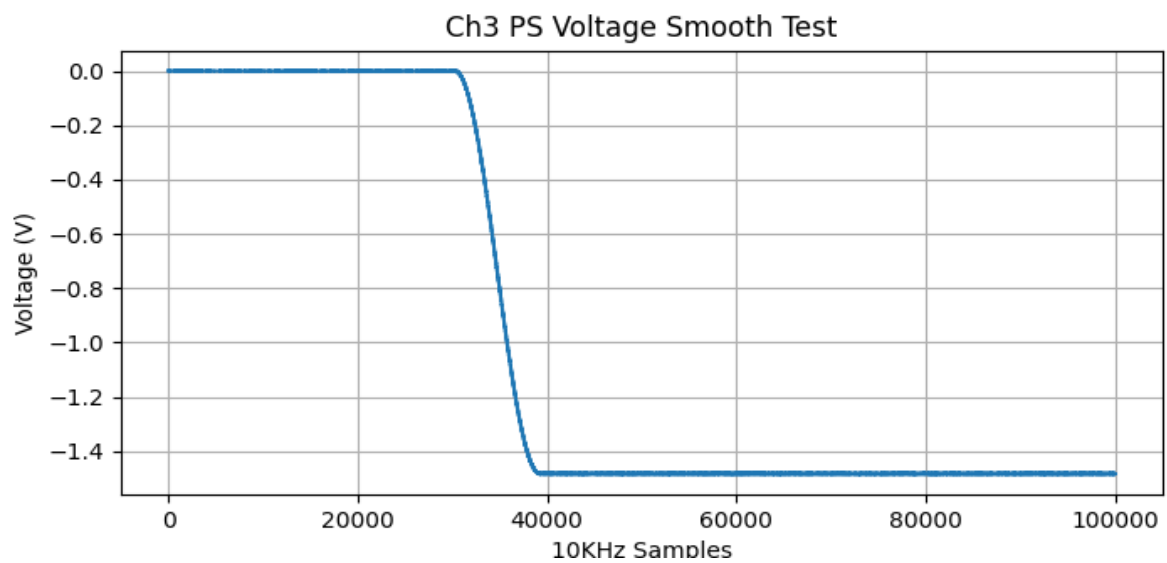
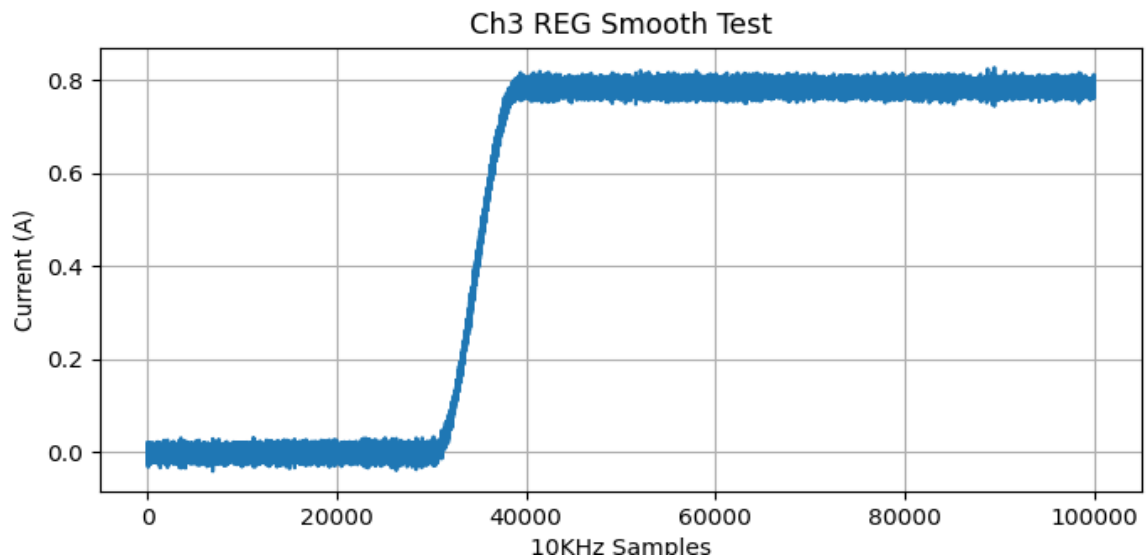
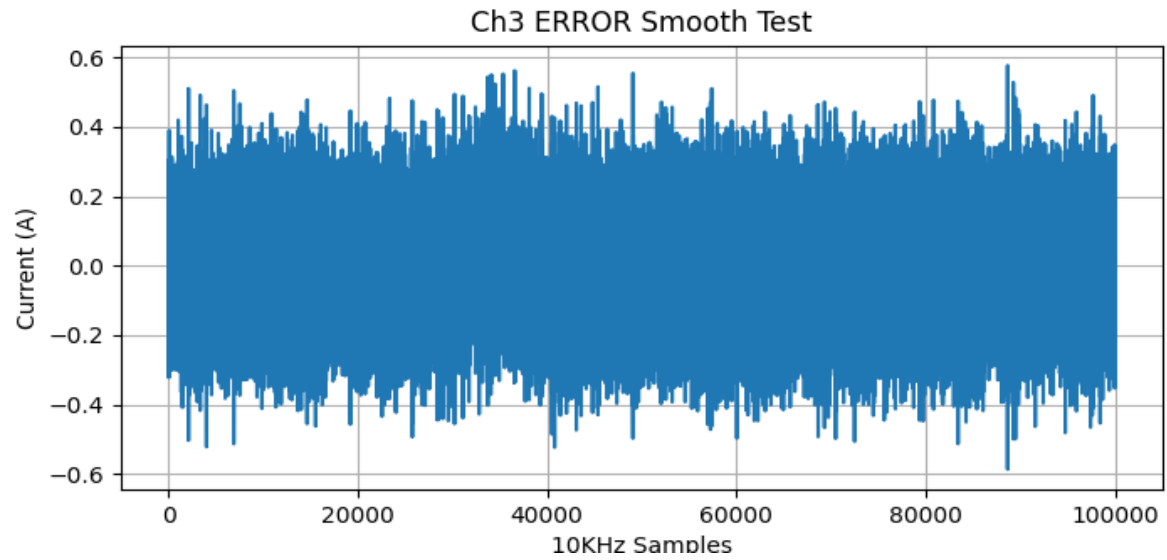
Jump Test Results: CH3



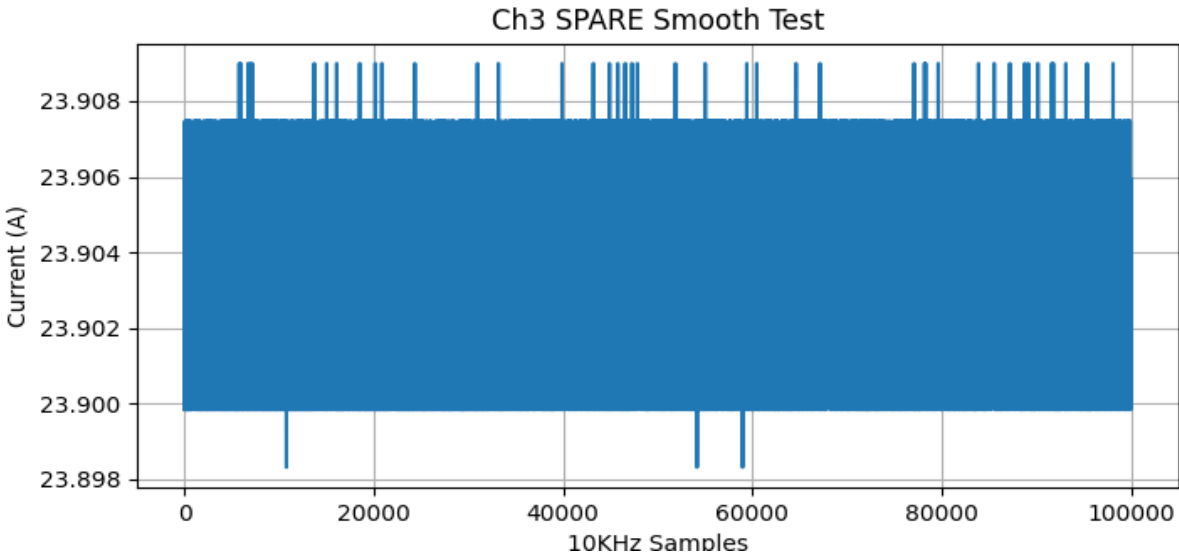
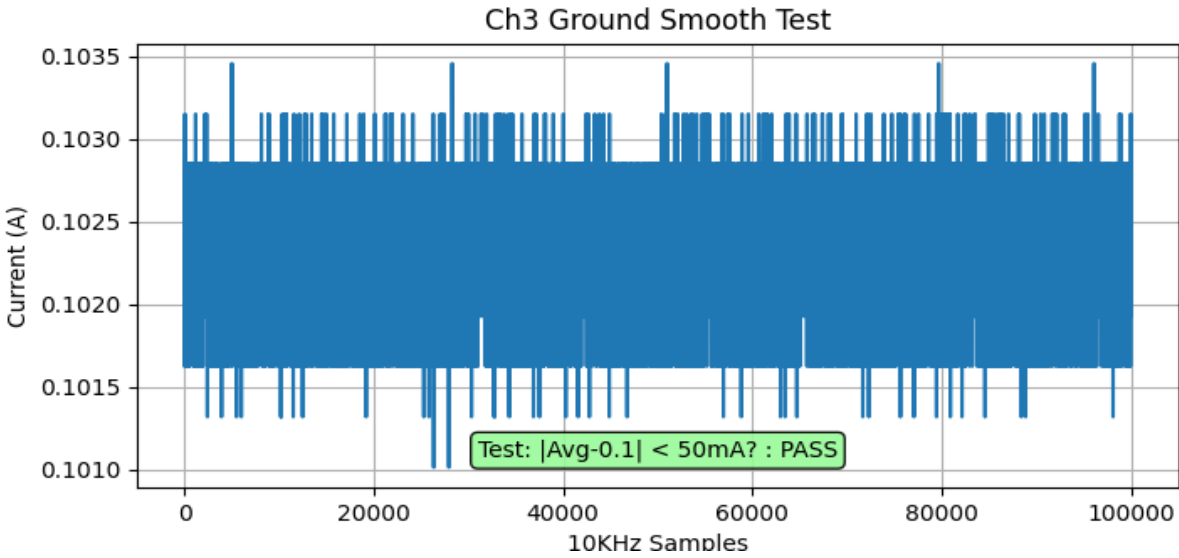
Smooth Test Results: CH3



Smooth Test Results: CH3



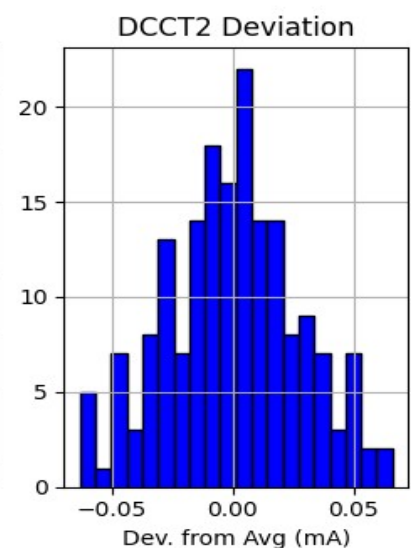
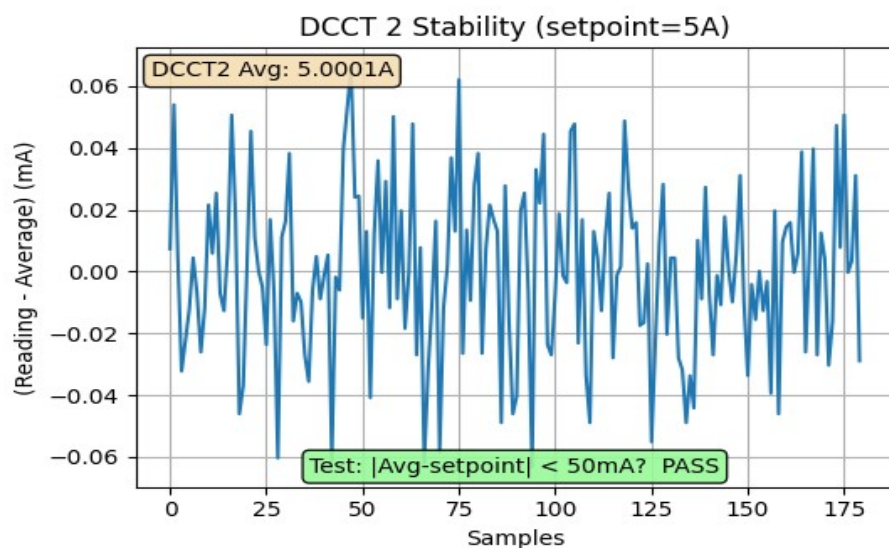
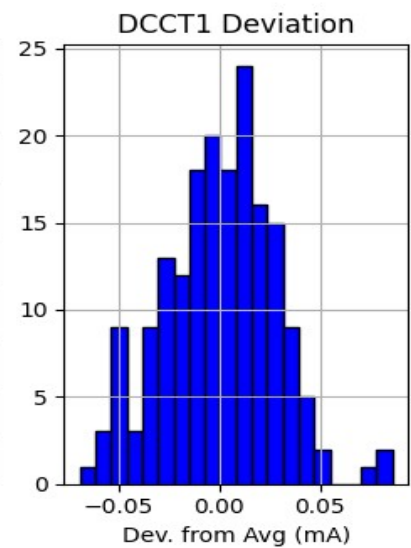
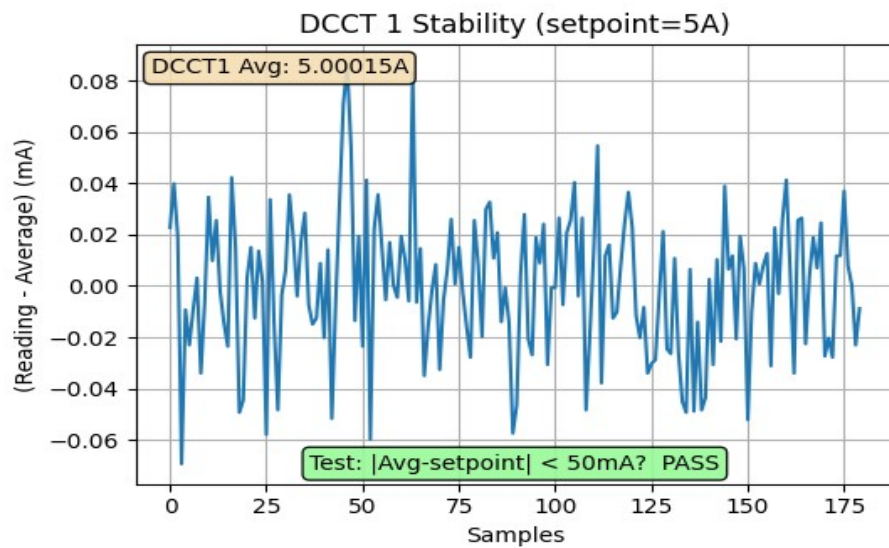
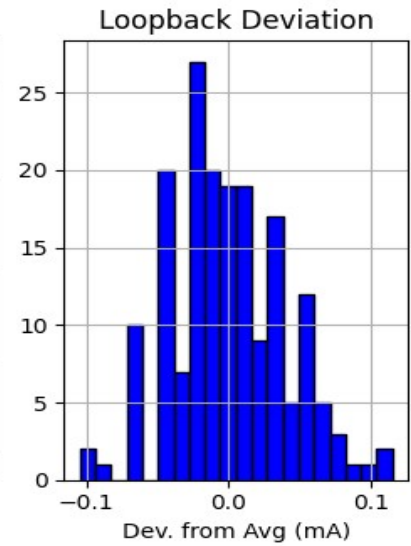
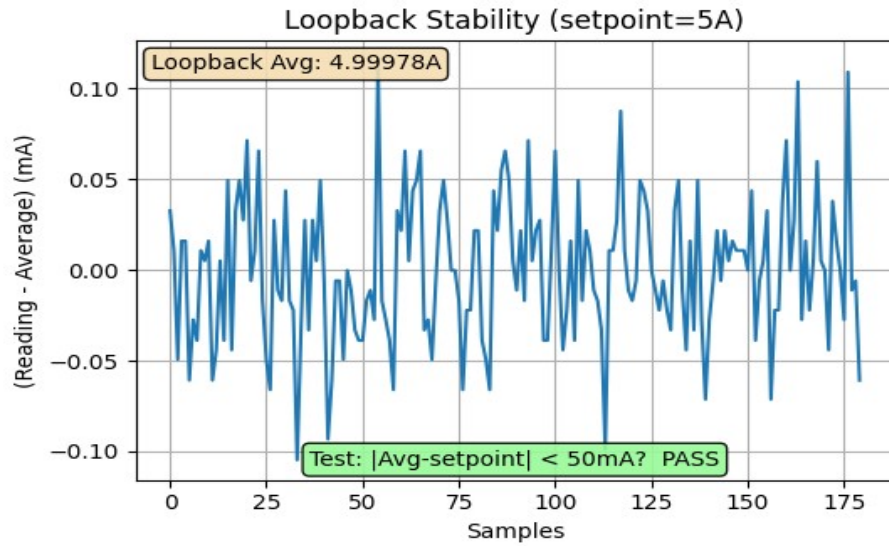
Smooth Test Results: CH3



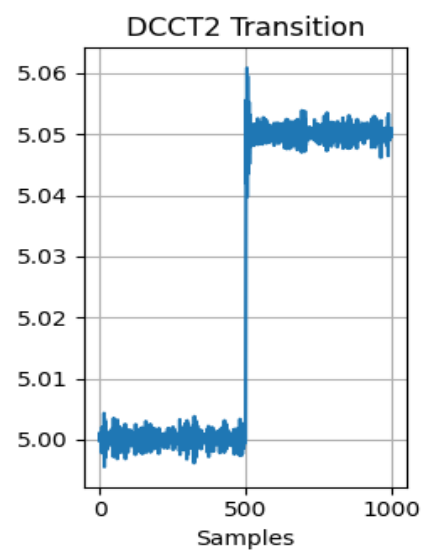
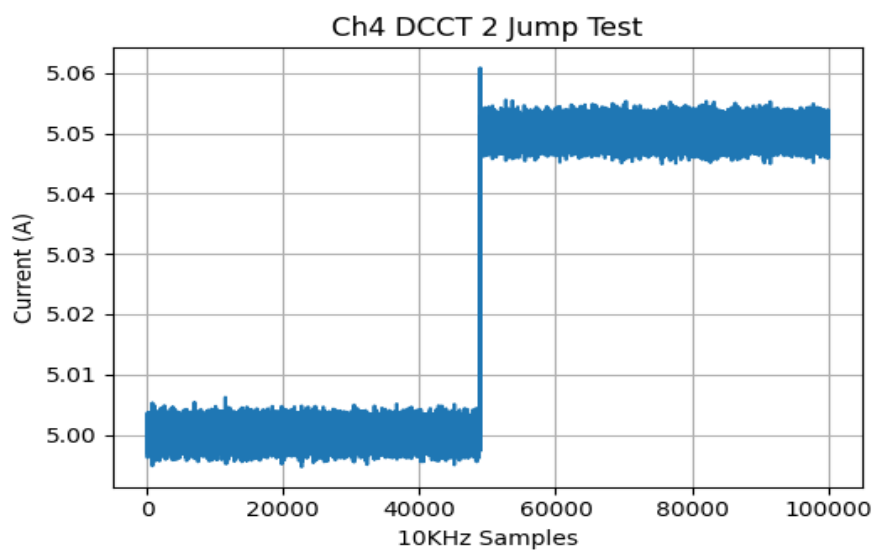
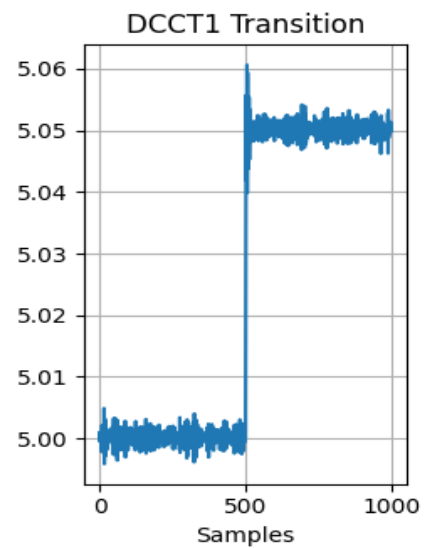
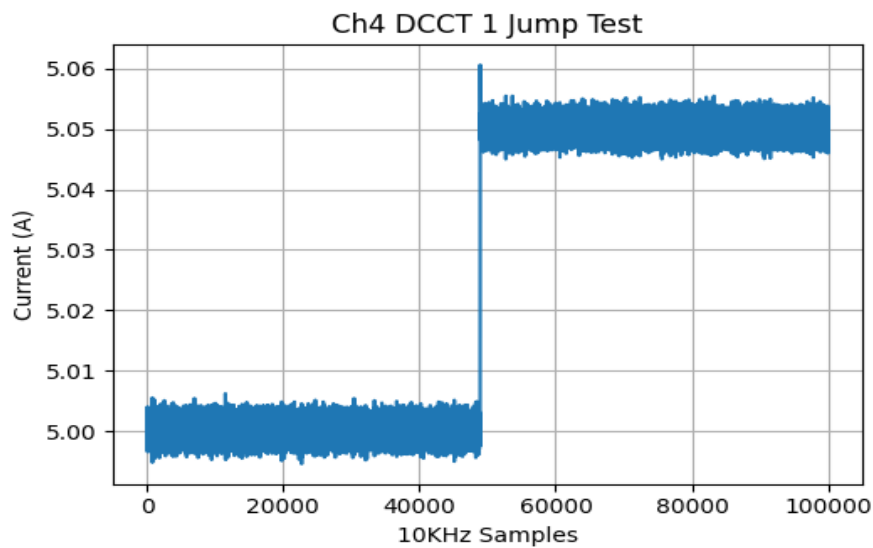
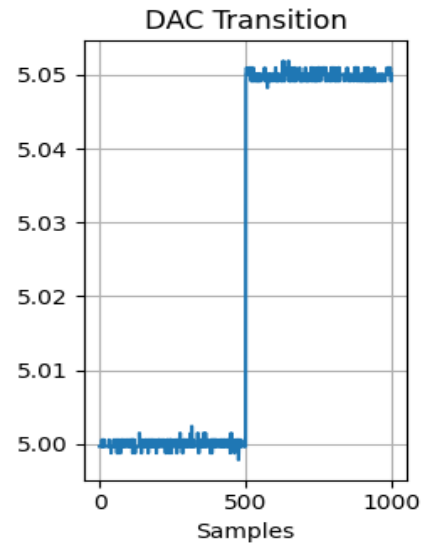
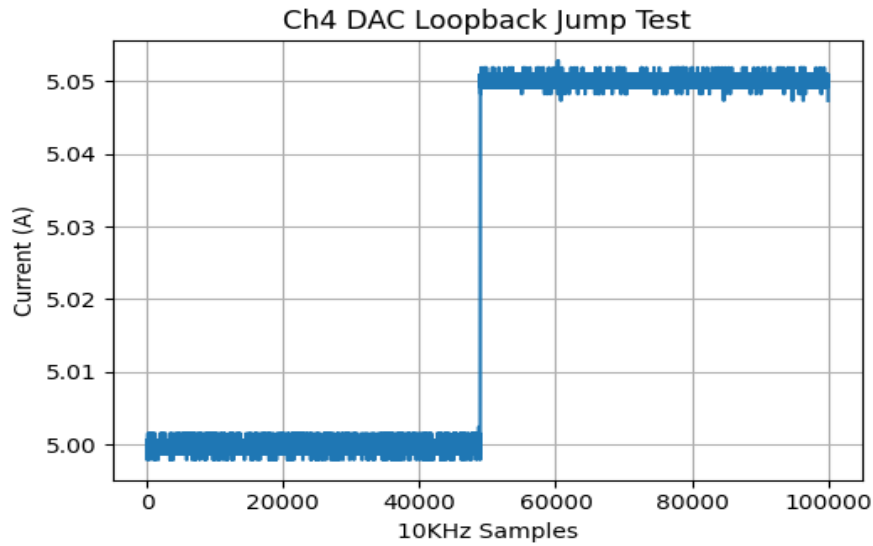
Channel 4

ATE Fault Tests for Channel 4	
Fault #1 Generated and Cleared	PASS
Fault #2 Generated and Cleared	PASS
Fault SPARE Generated and Cleared	PASS
Fault DCCT Generated and Cleared	PASS

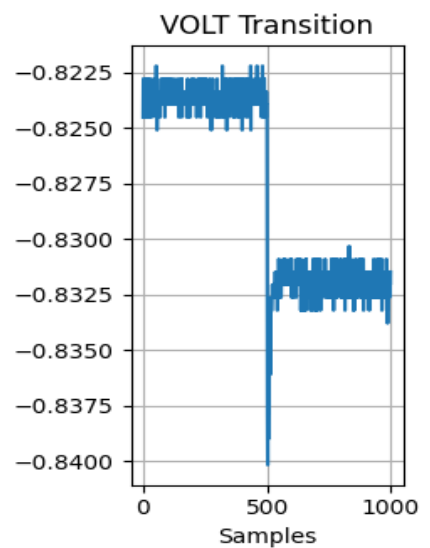
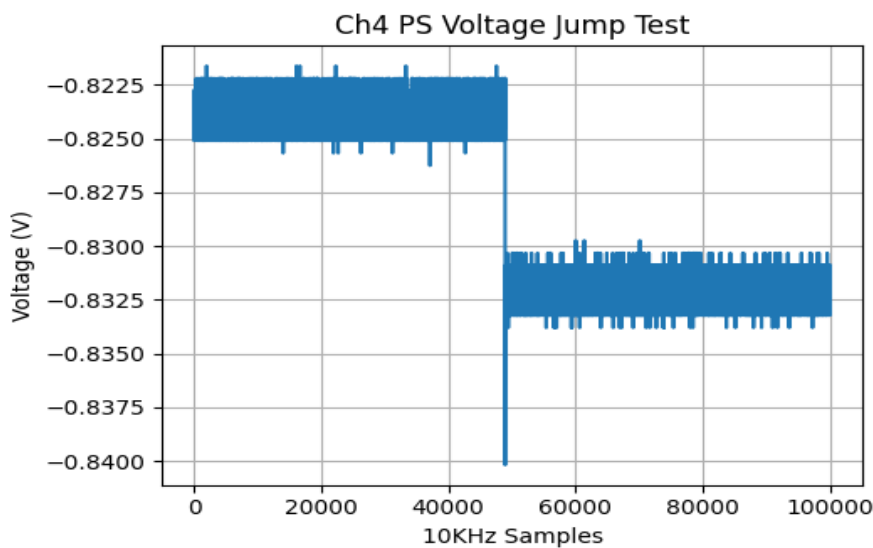
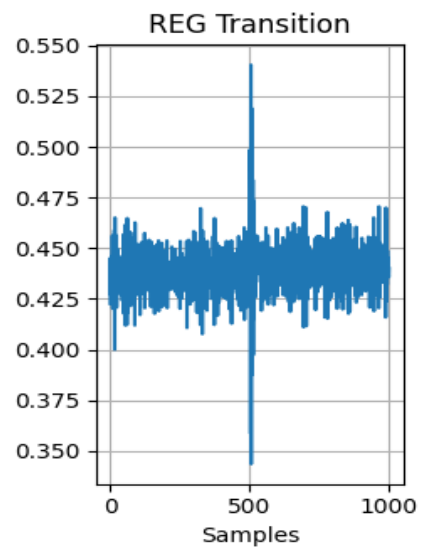
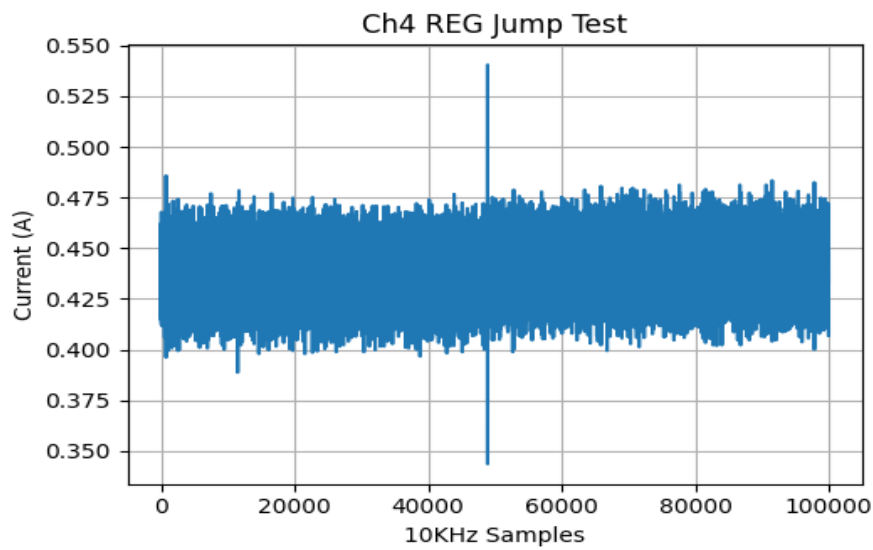
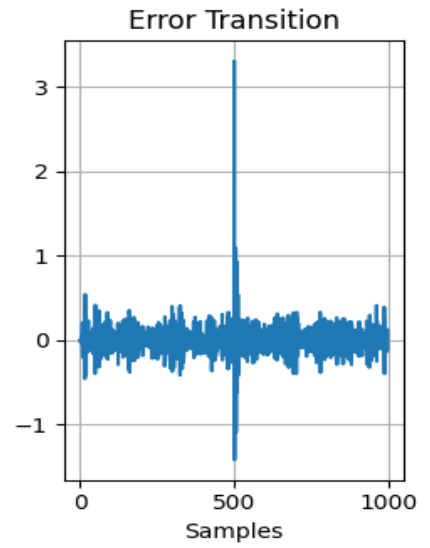
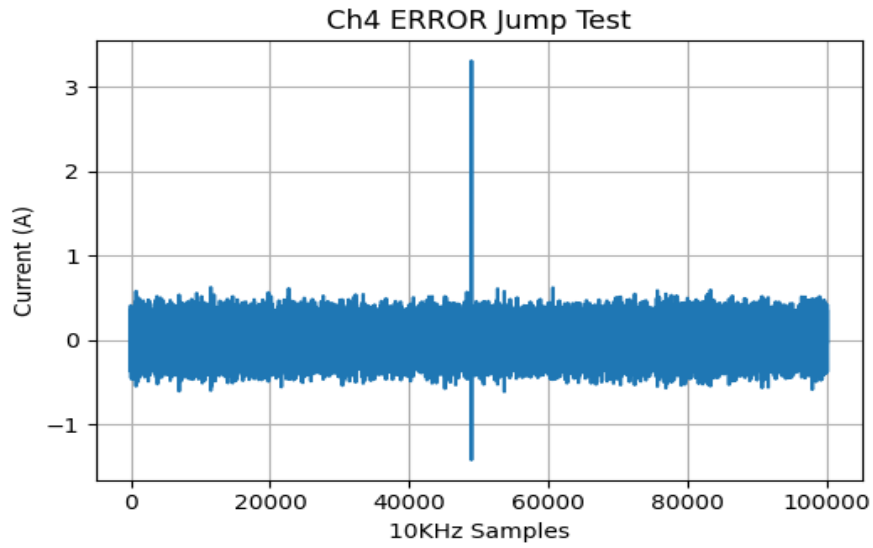
Power Supply Regulation for Channel 4:



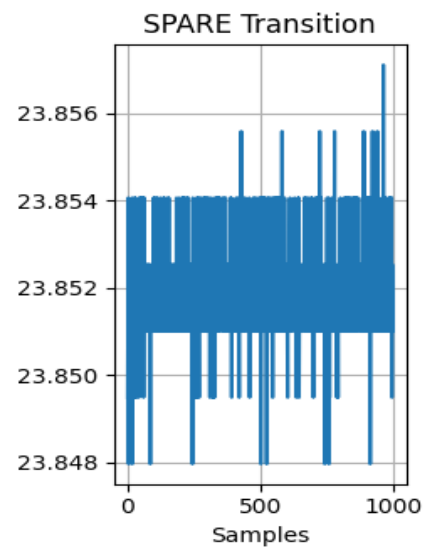
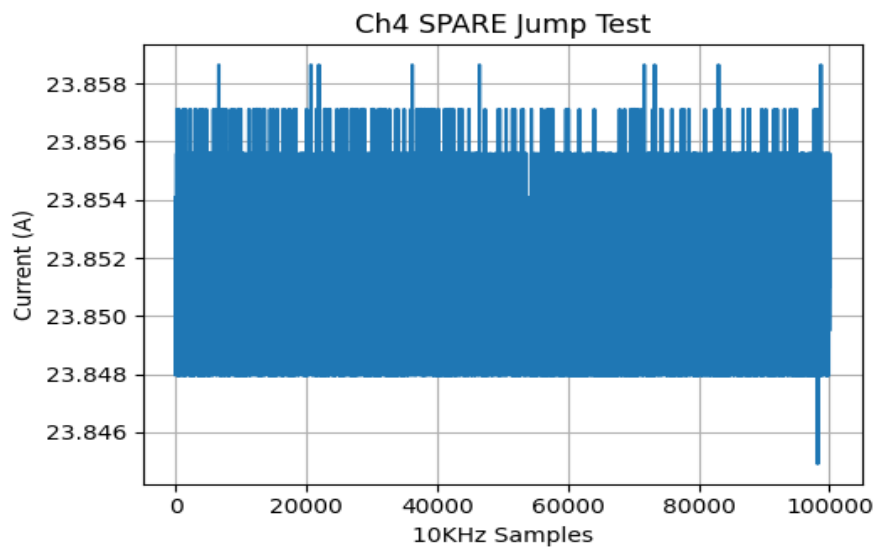
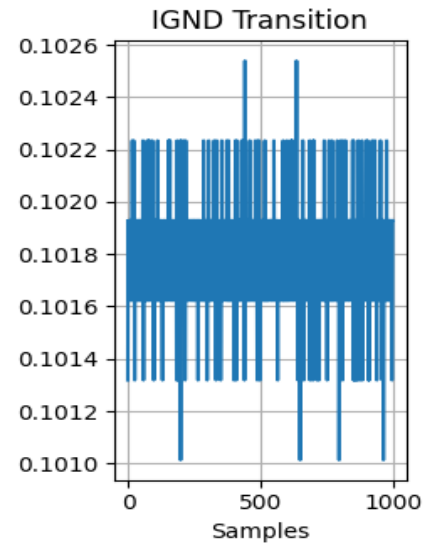
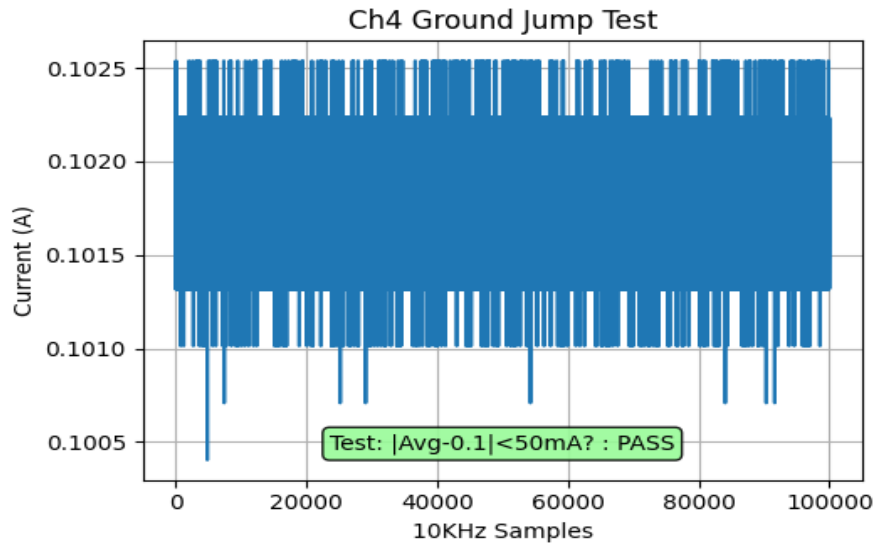
Jump Test Results: CH4



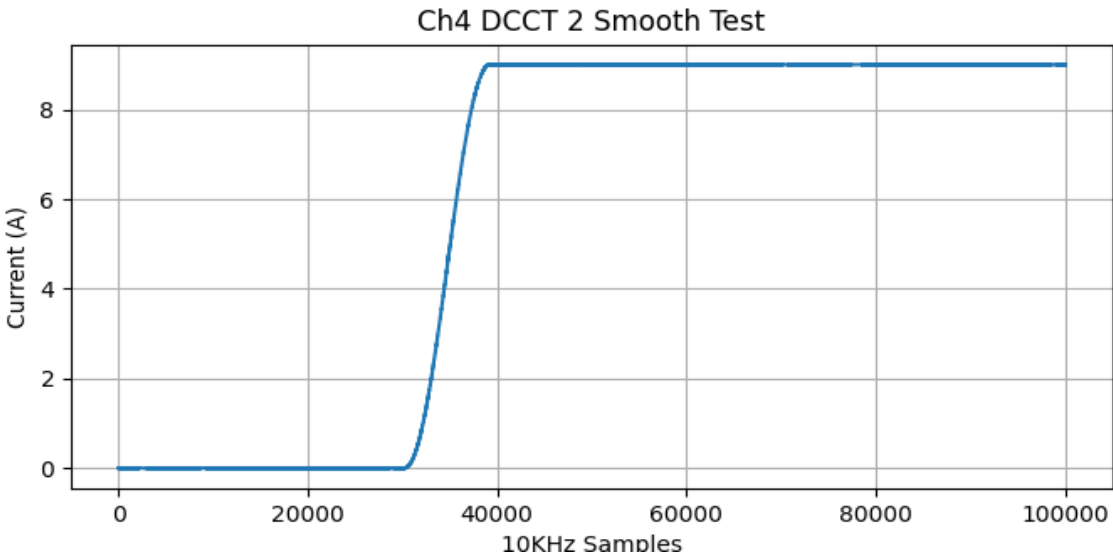
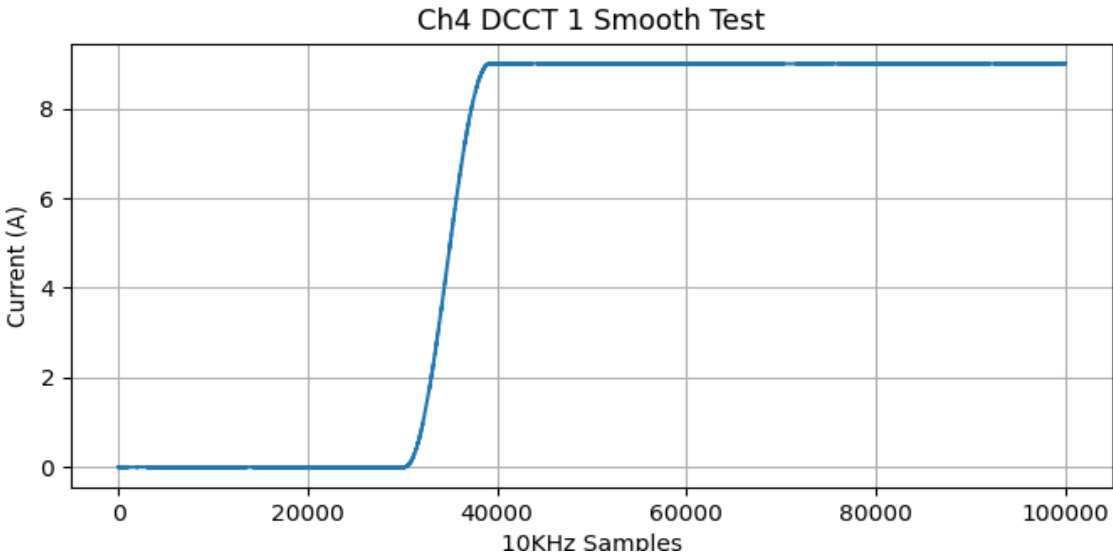
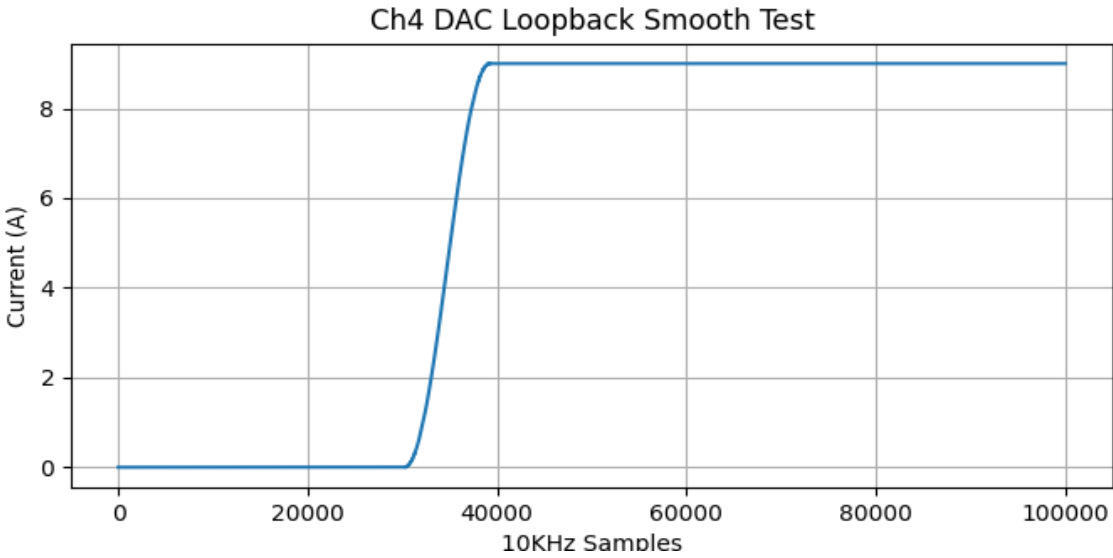
Jump Test Results: CH4



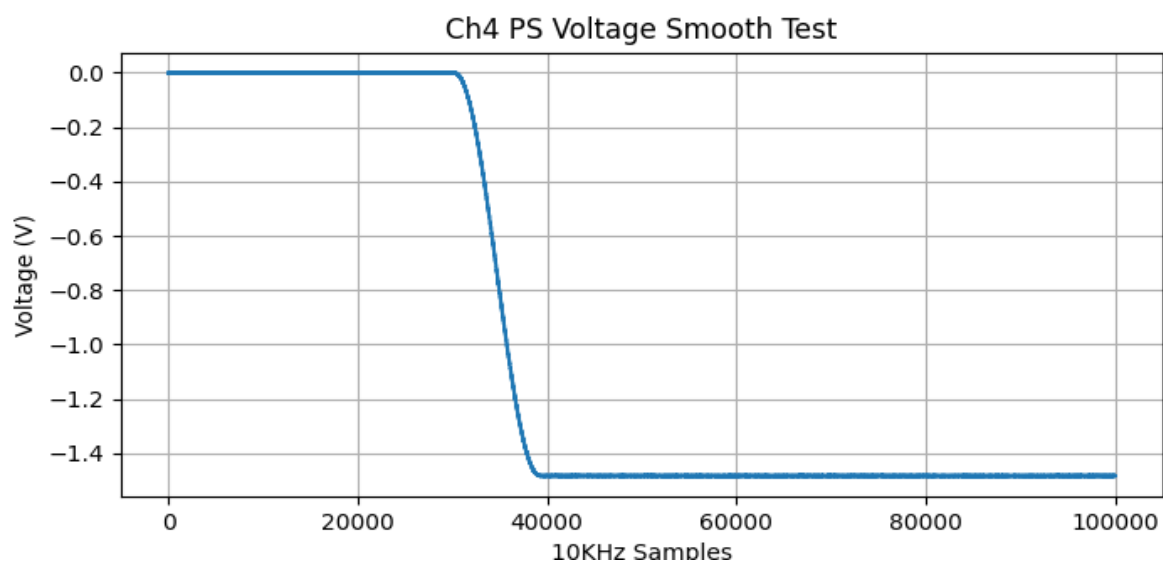
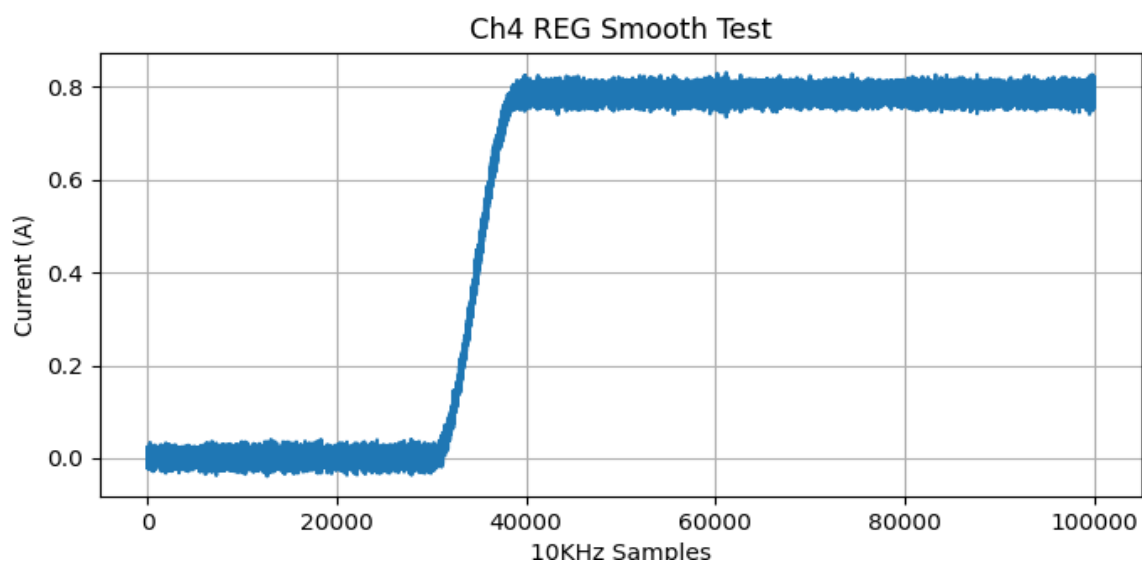
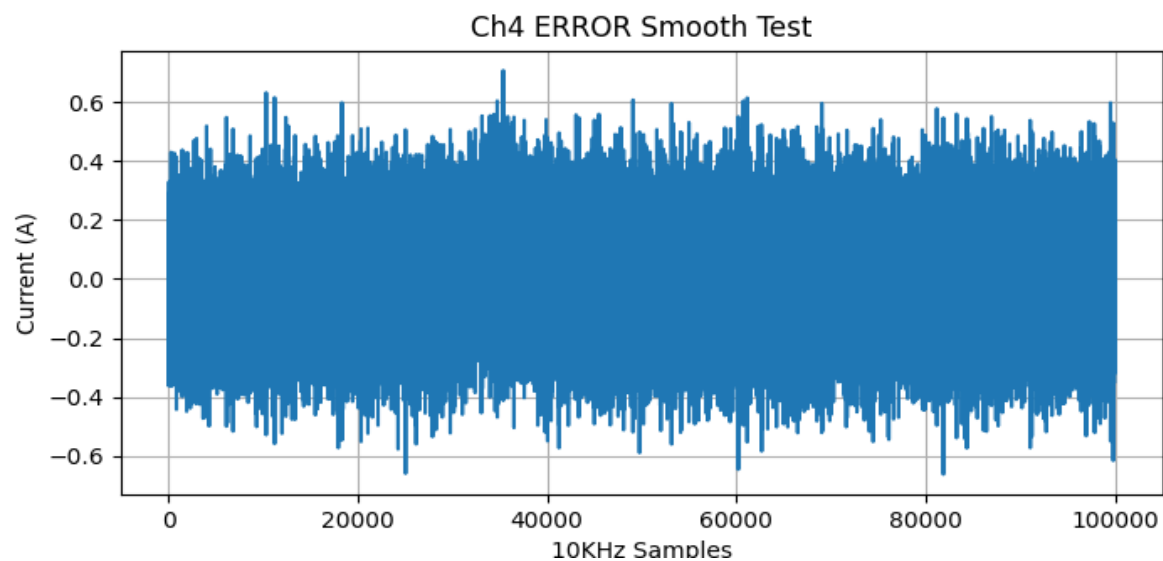
Jump Test Results: CH4



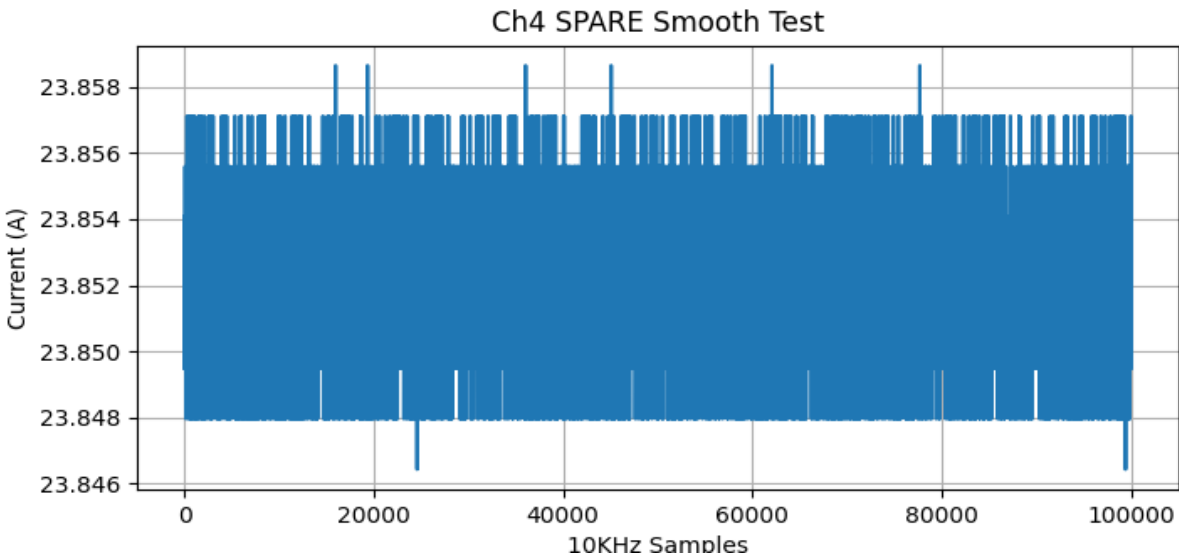
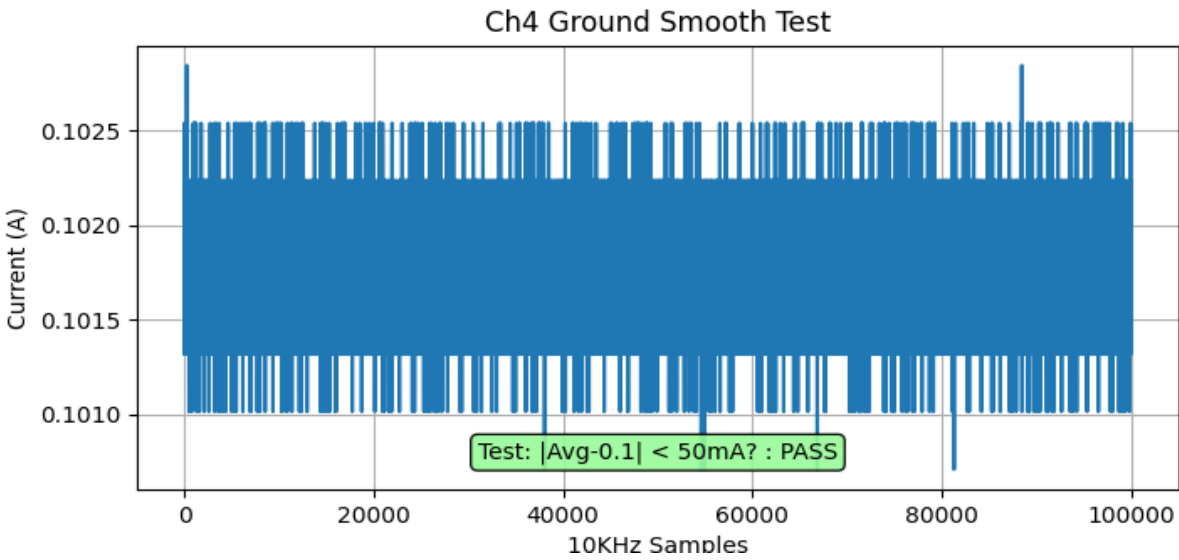
Smooth Test Results: CH4



Smooth Test Results: CH4



Smooth Test Results: CH4



FOFB TX test (Bandwidth-Mode = Fast)

PV	Value	Pass?
lab{2}Chan1:DAC-I	11.500301361083984	PASS
lab{2}Chan2:DAC-I	11.500070571899414	PASS
lab{2}Chan3:DAC-I	11.50058650970459	PASS
lab{2}Chan4:DAC-I	11.500749588012695	PASS

UDP RX Packet Test

```

14:49:17.193842 IP (tos 0x0, ttl 64, id 14145, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.51845 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.194500 IP (tos 0x0, ttl 64, id 14146, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.39279 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.205308 IP (tos 0x0, ttl 64, id 14151, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.35592 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.206124 IP (tos 0x0, ttl 64, id 14152, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.42118 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.217025 IP (tos 0x0, ttl 64, id 14159, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.59623 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.217908 IP (tos 0x0, ttl 64, id 14160, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.56165 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.228816 IP (tos 0x0, ttl 64, id 14171, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.49407 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.229579 IP (tos 0x0, ttl 64, id 14172, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.43055 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.240338 IP (tos 0x0, ttl 64, id 14182, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.58561 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.241085 IP (tos 0x0, ttl 64, id 14183, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.55181 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.252203 IP (tos 0x0, ttl 64, id 14192, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.51105 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.252926 IP (tos 0x0, ttl 64, id 14193, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.54549 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.264250 IP (tos 0x0, ttl 64, id 14201, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.39520 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.264801 IP (tos 0x0, ttl 64, id 14202, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.39352 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.276130 IP (tos 0x0, ttl 64, id 14211, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.53274 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.276450 IP (tos 0x0, ttl 64, id 14212, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.57279 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.288053 IP (tos 0x0, ttl 64, id 14215, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.53980 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.288056 IP (tos 0x0, ttl 64, id 14216, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.44709 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.299828 IP (tos 0x0, ttl 64, id 14220, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.52280 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.299830 IP (tos 0x0, ttl 64, id 14221, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.46414 > 10.69.26.55.12345: [udp sum ok] UDP, length 36

```

(Full tcpdump log saved to /home/pstester/NSLS-II_PSCs/NEXT-II/NSLS2_zPSC_Calibration_and_Test/test_data/raw_logs/4ch_HSF_SN0001_RawData_05-19-26_14-49/tcpdump_full.log)

Test	Result
UDP RX FOFB	PASS